

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**RESPONDENTS' IDENTIFICATION OF EXPERT WITNESSES
FOR CLAIM CONSTRUCTION**

Pursuant to Order No. 6, Setting the Procedural Schedule on June 9, 2026, Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively "Tower Semiconductor" or "Respondents") hereby identify the following expert witnesses, including their expertise and curriculum vitae, and whom may proffer opinions regarding claim construction issues for the asserted claims of the patents-in-suit in this investigation. Respondents may call to offer expert testimony or expert opinions at the hearing in the above-referenced investigation, and as such, Respondents reserve the right to rely upon testimony from any of the experts identified herein. Respondents reserve their rights to amend or supplement these disclosures as this investigation and discovery proceeds.

Subject to the above, Respondents identify the following expert witnesses for claim construction:

I. Frank Ernst, Ph.D.

Dr. Ernst is an expert in the technical matters of this case. Dr. Ernst's curriculum vitae, attached hereto as Exhibit A, sets forth his qualifications.

Dr. Ernst may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Ernst may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

II. David K. Choi, Ph.D.

Dr. Choi is an expert in the technical matters of this case. Dr. Choi's curriculum vitae, attached hereto as Exhibit B, sets forth his qualifications.

Dr. Choi may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to

satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Choi may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

III. Jeffrey Bokor, Ph.D.

Dr. Bokor is an expert in the technical matters of this case. Dr. Bokor's curriculum vitae, attached hereto as Exhibit C, sets forth his qualifications.

Dr. Bokor may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Bokor may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

IV. Sanjay K. Banerjee, Ph.D.

Dr. Banerjee is an expert in the technical matters of this case. Dr. Banerjee's curriculum vitae, attached hereto as Exhibit D, sets forth his qualifications.

Dr. Banerjee may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted

Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Banerjee may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

V. Suman Datta, Ph.D.

Dr. Datta is an expert in the technical matters of this case. Dr. Datta's curriculum vitae, attached hereto as Exhibit E, sets forth his qualifications.

Dr. Datta may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Datta may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

Date: August 17, 2026

/s/ Stephen R. Smith

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CERTIFICATE OF SERVICE

I hereby certify that on August 17, 2026, copies of the foregoing were filed with and served upon the following as indicated:

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EXHIBIT A

Curriculum Vitae

Frank Ernst

Dr. rer. nat. habil.

Leonard Case Jr. Professor of Engineering

Department of Materials Science and Engineering

Case Western Reserve University

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August 3, 2026

Academic Degrees

1997	Dr. rer. nat. habil. (Habilitation*)	Materials Science	University of Stuttgart, Germany (Advisors: E. Arzt, F. Aldinger, M. Rühle)
1987	Dr. rer. nat. (Ph. D.)	Physical Metallurgy	University of Göttingen, Germany (Advisor: P. Haasen)
1984	Dipl.-Phys. (Masters)	Physics	University of Göttingen, Germany (Advisor: P. Haasen)

Bio Data

Date of Birth	November 17, 1959
Place of Birth	Göttingen, Germany
Spouse	Sabine Ernst
Children	Meike Ernst (October 20, 1990) Oliver Ernst (September 28, 1992)

*The Habilitation is the highest academic degree in Germany. It is a post-doctoral qualification awarded for demonstrating a special qualification in research and *teaching* at the university level.

**The *venia legendi* is the permission to teach at a German university. It *can* be granted on the basis of a Habilitation.

Appointments

2016–2024	Department Chair Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
2016–2021	Faculty Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2012	Secondary Professor Department of Chemical and Biomolecular Engineering CWRU, Cleveland, (Ohio), USA
2008–2011	Director Case Center for Surface Engineering CWRU, Cleveland, (Ohio), USA
2008–2016	Co-Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2007	Leonard Case Jr. Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
Since 2000	Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
1997–2000	Lecturer University of Stuttgart, Germany
1989–2000	Senior Scientist Max-Planck-Institut für Metallforschung Stuttgart, Germany
1987–1988	Post-Doctoral Researcher Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA

Awards

- 2025 **Nomination for John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2019 **Jessica Melton Perry Award for Disnguished Teaching in Disciplinary & Professional Writing**
CWRU
- 2019 **Graduate Teaching Award**
Case School of Engineering
- 2018 **Nomination for the Graduate Teaching Award**
Case School of Engineering
- 2017 **John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2016 **Technical Educator Award**
CTSC (Cleveland Technical Societies Council)
- 2010 **Acta Materialia Top Reviewer 2009**
Elsevier
- 2010 **Materials Science and Engineering Outstanding Reviewer 2009**
Elsevier
- 2009 **Materials Science and Engineering Outstanding Reviewer 2008**
Elsevier
- 2006 **Case School of Engineering Outstanding Teacher 2005-2006**
CWRU
- 2005 **Nomination for the
2005 Carl F. Wittke Award for Excellence in Undergraduate Teaching**
CWRU
- 2005 **2004-2005 Case School of Engineering Teaching Leader**
CWRU
- 2004 **Tenure Appointment**
CWRU
- 2001 **Werner Köster Prize 2000**
German Materials Research Society (DGM) and Zeitschrift für Metallkunde
- 1997 **Venia Legendi****
University of Stuttgart
- 1993 **Tenure Appointment**
Max-Planck Society, Max-Planck-Institut für Metallforschung
- 1985 **Graduate Student Research Award**
German Materials Research Society (DGM)
- 1984 **Heineman Scholarship for Graduate Studies**
Minna-James-Heineman-Foundation (for 3 years)

Professional Activities

Professional Memberships

ASM	ASM International
MSNO	Microscopy Society of Northern Ohio
MSA	Microscopy Society of America
ACerS	American Ceramic Society
TMS	The Minerals, Metals & Materials Society
DGE	German Electron Microscopy Society
DGM	German Materials Research Society

Memberships on Boards of Scientific Journals

2010 – today	Metals: Editorial Board
2006 – 2016	International Journal of Materials Research: Advisory Board
2004 - 2013	Materials Science and Engineering A: Editorial Board
1997 – 2006	Zeitschrift für Metallkunde: Advisory Board
1996 – 2002	Journal of Electron Microscopy: Editorial Board

Service as Reviewer

For Scientific Publications

Acta Crystallographica	Acta Materialia
Advanced Materials	Applied Physics Letters
International Journal of Materials Research	Journal of Biomedical Materials Research
Journal of Electron Microscopy	Journal of Physical Chemistry
Journal of Materials Research	Materials Science and Engineering
Materials Science and Technology	Metallurgical and Materials Transactions
Metals	Microscopy & Microanalysis
Nano Letters	Philosophical Magazine
Physica Status Solidi	Physical Review B
Physical Review Letters	Science
Steel Research	Surface Science
Surface Science and Technology	Ultramicroscopy
Wiley & Sons	Zeitschrift für Metallkunde

For Research Agencies

European Science Foundation
German Science Foundation (Research Proposals, Proposals for Research Centers)
Max-Planck-Society (Proposals for International Max-Planck Research Schools)
National Science Foundation
US Department of Defense

Conference Activity

1. Organizer of "Third Symposium on Surface Hardening of Corrosion-Resistant Alloys" *Materials Science & Technology 2014*. Pittsburgh, PA, USA, October 12-16, 2014.
2. Session Chair of Symposium "Grain Boundary Engineering" *XIII International Conference on Intergranular and Interface Boundaries (iib2010)*. Mie, Japan, June 27 to July 2, 2010.
3. Co-Organizer and Session Chair of the "International Symposium on Surface Hardening of Corrosion Resistant Alloys". Cleveland, May 24 to 26, 2010.
4. Co-Organizer and Session Chair of the "International Meeting on Surface Hardening of Stainless Steel". Cleveland, October 22 and 23, 2007.
5. Co-Organizer and Session Chairs of Conference on "Progress and Challenges in Understanding Materials Interfaces." *Material Science and Technology Conference and Exhibition*. Cincinnati, October 15-19, 2006.
6. Organizer and Chair of "Interfaces: A Symposium in Honor of Manfred Rühle." *Annual Meeting of the Microscopy Society of America*. Savannah, Georgia, USA, August 01-05, 2004.
7. Chair of the Symposium "Materials Design." *First Kyoto Workshop on Computational Materials Science*. Kyoto, Japan, March 4, 2003.
8. Organizer and Chair of the *MSNO Winter Symposium on Transmission Electron Microscopy*. Cleveland, Ohio, USA, February 13, 2002.
9. Chair of a Symposium on "Data Analysis and Modeling Across length Scales." *United Engineering Foundation Meeting on Integrating Materials Science into Engineering Structures and Devices*. Lake Arrowhead, California, USA, November 11-15, 2001.
10. Chair of a Symposium on "Quantitative Analysis of Interface Structure by HRTEM." *Annual Meeting of the Microscopy Society of America 2001*. Long Beach, California, USA, August 05-09, 2001.
11. Chair of a Symposium on "Wide Band-Gap Semiconductors." *Microscopy of Semiconducting Materials XII*. Oxford, United Kingdom, March 25-29, 2001.
12. Organizer and Chair of a Symposium on "Quantitative High-Resolution Electron Microscopy." *EUREM 12*. Brno, Czech Republic, July 9-14, 2000.

13. Organizer and Chair of an *International Conference on Quantitative High-Resolution Electron Microscopy*. Berlin, Germany, March 27-29, 2000.
14. Organizer and Chair of a Symposium on "Materials Research." *29th Congress of the German Electron Microscopy Society*. Dortmund, Germany, September 5-10, 1999.
15. Organizer and Chair of a Symposium on "Advances in Quantitative and Coherent High-Resolution Electron Microscopy." *14th International Congress on Electron Microscopy*. Cancun, Mexico, August 31 - September 4, 1998.
16. Organizer and Chair of the International Conference on *Frontiers of Electron Microscopy in Materials Science*. Irsee, Germany, April 19-24, 1998.
17. Organizer and Chair of a Symposium on "Interfaces and Surfaces." *Multinational Congress of the Austrian, German, and Swiss Microscopy Societies*. Regensburg, Germany, September 8-12, 1997.
18. Chair of a Symposium on "Atomic Resolution Microscopy of Surfaces and Interfaces." *Fall Meeting of the Materials Research Society*. Boston, USA, December 2-6, 1996.
19. Chair of a Symposium on "Defects at Hetero-Interfaces." *Annual Meeting of The Materials Society*. San Francisco, USA, February 27-March 3, 1994.
20. Chair of a Symposium on "Defect-Interface Interactions." *Fall Meeting of the Materials Research Society*. Boston, USA, November 29-December 3, 1993.
21. Chair of a Symposium on "Stress Relaxation and Misfit Accommodation in Epitaxial Thin Films." *Annual Meeting of The Materials Society*. San Francisco, USA, February 17-22, 1991.
22. Chair of a Symposium on "HRTEM of Defects in Materials." *Spring Meeting of The Materials Research Society*. San Francisco, USA, April 16-20, 1990.

Publications

Academic Publications

1. F. Ernst: *Struktur von Heterogrenzflächen (Structure of Heterointerfaces)*. Habilitationsschrift (Professorial Thesis), University of Stuttgart, 1997.
2. F. Ernst: *Hochauflösende Elektronenmikroskopie der Entmischung von Al-1 at% Ag at 413 K (Decomposition Kinetics of Al-1 at% Ag at 413 K, Studied by High-Resolution Electron Microscopy)*. Dissertation (Ph. D. Thesis), University of Göttingen, 1987.
3. F. Ernst: *Rekristallisationsexperimente in $\langle 100 \rangle$ -zugverformten Kupfer-Einkristallen mit künstlicher Keimbildung (Recrystallization of $\langle 100 \rangle$ -Strained Copper Single Crystals After Artificial Nucleation)*. Diplomarbeit (Masters Thesis), University of Göttingen, 1984.

Refereed Publications[§]

1. Dernek, Ozan and **Mehdi, Redad** and Yue, Weiqi and Bachman, Jonah A. and Holt, Finley R. and Ponon, Gabriel O. and Tripathi, Pawan K. and Willard, Matthew A. and Ernst, Frank and French, Roger H.: *A Perspective on Synchrotron Data Science*. Interdisciplinary Information Sciences, vol. 31, pp. 83–88, 2025.
2. **Redad Mehdi**, Rounak Chawla, Erika I. Barcelos, Roger H. French, Frank Ernst: *2D-diffractogram Analysis: Kinematic-diffraction Simulator for Neural-network Training-data Generation*. Computational Materials Science, vol. 252, pp. 113777, 2025.
3. **Sameera Nalin Venkat**, Thomas G Ciardi, Mingjian Lu, Preston C DeLeo, Jube Augustino, Adam Goodman, Jayvic Cristian Jimenez, Anirban Mondal, Frank Ernst, Christine A Orme, Yinghui Wu, Roger H French, Laura S Bruckman: *A General Materials Data Science Framework for Quantitative 2D Analysis of Particle Growth from Image Sequences*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 71-82, 2024.
4. **Weiqi Yue**, Pawan K. Tripathi, Gabriel Ponon, Zhuldyz Ualikhankyzy, Donald W. Brown, Bjorn Clausen, Maria Strantza, Darren C. Pagan, Matthew A. Willard, Frank Ernst, Erman Ayday, Vipin Chaudhary, Roger H. French: *Phase Identification in Synchrotron X-ray Diffraction Patterns of Ti-6Al-4V Using Computer Vision and Deep Learning*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 36-52, 2024.

[§]Author names in **boldface** indicate students or post-doctoral scholars I advised.

5. **WeiQi Yue, Mohommad Redad Mehdi**, Pawan K Tripathi Matthew A. Willard, Frank Ernst, Roger H. French: *Exploring 2d X-ray Diffraction Phase Fraction Analysis With Convolutional Neural Networks: Insights From Kinematic-diffraction Simulations*. MRS Advances, vol. 9, pp. 921-928, 2024.
6. Leveraging Deep Neural Networks for Feature Extraction from 2D XRD Diffractograms Acquired at Synchrotron Beamlines: *Pawan K. Tripathi, WeiQi Yue, Redad Mehdi, Finley Holt, Gabriel Ponon, Vipin Chaudhary, Frank Ernst, Matthew Willard, Bjorn Clausen Donald W. Brown, Daniel Savage, Roger H. French*. Los Alamos National Laboratory (LANL) Report, vol. LA-UR-23-33318, pp. 1, 2023.
7. **C. Illing, Z. Ren, F. Ernst**: *Low-Temperature Carburization: Ex Situ Activation of Austenitic Stainless Steel*. Metals, vol. 2023, pp. 335-359, 13.
8. **C. Illing, M. Bestic, F. Ernst**: *Additive Manufacturing: Corrosion Proofing by Infusion of Interstitial Solute—Exemplified for Alloy 22*. Metals, vol. 2023, pp. 127-143, 13.
9. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the efficacy of concentrated interstitial carbon in enhancing the pitting corrosion resistance of stainless steel; response to Comment by Martin et al.*. Scripta Materialia, vol. 224, pp. 115084, 2023.
10. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the Efficacy of Concentrated Interstitial Carbon in Enhancing the Pitting Corrosion Resistance of Stainless Steel*. Acta Materialia, vol. 221, pp. 117433-447, 2021.
11. **C. Illing, Z. Ren, A. Agaponova**, A. Heuer, F. Ernst: *Rapid Alloy Surface Engineering Through Closed-Vessel Reagent Pyrolysis*. Metals, vol. 11, pp. 1764-1782, 2021.
12. **Z. Ren**, F. Ernst: *Stress–Corrosion Cracking of AISI 316 L Stainless Steel in Seawater Environments: Effect of Surface Machining*. Metals, vol. 10, pp. 1324-1341, 2020.
13. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *The Passivity of Low-Temperature Carburized Austenitic Stainless Steel AISI-316L in a Simulated Boiling-Water-Reactor Environment*. Journal of Nuclear Materials, vol. 537, pp. 152197, 2020.
14. **Z. Ren**, F. Ernst: *High-Temperature Phase Transformations in AISI 316 Stainless Steel Infused With Concentrated Interstitial Carbon*. Journal of Alloys and Compounds, vol. 819, pp. 153000, 2020.

15. **J.C. Dalton**, F. Ernst, A.H. Heuer: *Low-Temperature Nitridation of 2205 Duplex Stainless Steel*. Metallurgical and Materials Transactions A, vol. 51, pp. 607-617, 2020.
16. **Z. Ren**, F. Ernst: *Electronic Impact of Concentrated Interstitial Carbon on Physical Properties of AISI-316 Austenitic Stainless Steel*. Acta Materialia, vol. 2019, pp. 96-105, 173.
17. **Z. Ren**, A.H. Heuer, F. Ernst: *Ultrahigh-Strength AISI-316 Austenitic Stainless Steel Foils Through Concentrated Interstitial Carbon*. Acta Materialia, vol. 2019, pp. 231-240, 167.
18. **Z. Ren**, S.Eppell, S. Collins, F. Ernst: *Co-Cr-Mo Alloys: Improved Wear Resistance Through Low-Temperature Gas-Phase Nitro-Carburization*. Surface & Coatings Technology, vol. 378, pp. 124943-1-9, 2019.
19. **X. Zhan**, M. Daniil, X. Liu, M.A. Willard, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Magnetic Properties of Amorphous Ni-P*. Journal of Alloys and Compounds, vol. 2019, pp. 742-749, 786.
20. **Z. Li, C. Illing**, A.H. Heuer, F. Ernst: *Low-Temperature Carburization of AL-6XN Enabled by Provisional Passivation Metals* 2018 997-1011
21. **Z. Li, X. Zhan**, T. Paramasivam, W. Niu, A.H. Heuer, S. Maloy, R. Lillard, F. Ernst: *Surface Engineering of IN-718 by Low-Temperature Carburization: Properties and Thermal Stability*. Surface Engineering (2018). 35 281-293
22. **S. Lingala**, D. Jayne, F. Ernst: *Effect of Lubricant Additives on the Tribological Behavior of Aluminum Alloy Against Steel*. International Journal of Materials Research (2018). 789-802
23. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *Passivity of Surface-Engineered Alloys in a Simulated Boiling-Water-Reactor Environment*. Corrosion, vol. 74, pp. 635-653, 2018.
24. **X. Zhan**, P. Zhang, P.M. Voyles, X. Liu, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Short-to-Medium-Range-Order Evolution and Crystallization Behavior of Near-Eutectic Amorphous Ni-P*. Acta Materialia, vol. 122, pp. 400-411, 2017.
25. **D. Wang**, H. Kahn, F. Ernst, A.H. Heuer: *“Colossal” Interstitial Supersaturation in Delta Ferrite in Stainless Steels: (II) Low-temperature Nitridation of the 17-7 PH Alloy*. Acta Materialia, vol. 124, pp. 237-246, 2017.
26. J. Wang, **Z. Li, D Wang**, S. Qiu, F. Ernst: *Thermal stability of low-temperature-carburized austenitic stainless steel*. Acta Materialia, vol. 128, pp. 235-240, 2017.

27. **A. Zangiabadi, J. Dalton, D. Wang, F. Ernst, A.H. Heuer:** *The Formation of Martensitic Austenite during Nitridation of Stainless Steel.* Metallurgical and Materials Transactions A, vol. 48A, pp. 8-13, 2017.
28. **F. Ernst:** *Background of SAM Atom-Fraction Profiles.* Materials Characterization, vol. 125, pp. 142-151, 2017.
29. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *“Colossal” Interstitial Supersaturation in Delta Ferrite in 17-7 PH Stainless Steels After Low-temperature Nitridation.* Microscopy and Microanalysis (22). 2016 2020-2023
30. **L. Yu, Z. Vashaei, F. Ernst, A. Kumar, R. Akolkar:** *Direct Electroless Deposition of Nano-scale Copper Films on Ruthenium: Role of Ethylenediamine Additive as Nucleation Promoter.* Journal of the Electrochemical Society, vol. 163, pp. D374-D378, 2015.
31. **X. Zhan and F. Ernst:** *Crystallization Micro-Mechanism of Near-Eutectic Amorphous Ni-P.* Acta Materialia, vol. 104, pp. 274-282, 2016.
32. **H. Kahn, F. Ernst, A.H. Heuer:** *Increasing the Coefficient of Sliding Friction of NiCr at Low Loads by Interstitial Surface Hardening.* Wear, vol. 346-347, pp. 1-5, 2016.
33. **D. Wu, Y. Ge, H. Kahn, A.H. Heuer, F. Ernst:** *Diffusion Profiles after Nitrocarburizing Austenitic Stainless Steel.* Surface Coatings and Technology, vol. 279, pp. 180-185, 2015.
34. **W. Niu, R.S. Lillard, Z. Li, F. Ernst:** *The Mechanism of Passivity in Austenitic Stainless Steel AISI 316L Carburized to Colossal Super Saturation.* Electrochimica Acta (176). 2015 410-419
35. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *NiAl Precipitation in Delta Ferrite Grains of 17-7 Precipitation-hardening Stainless Steel During Low-Temperature Interstitial Hardening.* Scripta Materialia, vol. 108, pp. 136-138, 2015.
36. **D. Wang, C.W. Chen, J.C. Dalton, F. Yang, R. Sharghi-Moshtaghin, H. Kahn, F. Ernst, R.E.A. Williams, D.W. McComb and A.H. Heuer:** *“Colossal” Interstitial Supersaturation of Delta Ferrite in Stainless Steels—I. Low-Temperature Carburization.* Acta Materialia, vol. 86, pp. 193-207, 2015.
37. **X. Zhan and F. Ernst:** *Crystallization of Ni-P Fabricated by Electroless Deposition: Microscopic Mechanism.* Materials Research Society Symposia Proceedings, vol. 1757, pp. mrsf14-1757-uu03-07, 2015.
38. **D. Wang, F. Ernst, H. Kahn, A.H. Heuer:** *Cellular Precipitation at a 17-7 PH Stainless Steel Interphase Interface during Low-Temperature Nitridation.* Metallurgical and Materials Transactions A (45). 2014 1-8

39. X. Gu, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Numerical Simulations of Carbon and Nitrogen Composition-Depth Profiles in Nitrocarburized Austenitic Stainless Steels*. Metallurgical and Materials Transactions, vol. 45A, pp. 4268-4279, 2014.
40. **X. Gu**, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Concentration Dependent Carbon Diffusivity in Austenite*. Metallurgical and Materials Transactions, vol. 45A, pp. 3790-3799, 2014.
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27. **R. Schweinfest**, F. Ernst, T. Wagner: *Struktur von Al/MgAl₂O₄-Grenzflächen*, in: *Werkstoffwoche '96, Symposium 7, Materialwissenschaftliche Grundlagen*, Eds. F. Aldinger, H. Mughrabi. DGM-Informationsgesellschaft mbH (Frankfurt), pp. 849-852, 1997.
28. **O. Kienzle**, F. Ernst, M. Rühle: *Struktur der $\Sigma 3$ (111) Kippkorngrenze in Strontiumtitanat*, in: *Werkstoffwoche '96, Symposium 7, Materialwissenschaftliche Grundlagen*, Eds. F. Aldinger, H. Mughrabi. DGM-Informationsgesellschaft mbH (Frankfurt), pp. 813-818, 1997.
29. F. Phillipp, F. Ernst: *High-Voltage Atomic Resolution Microscopy with the Stuttgart ARM 1250*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin 1996, CD-ROM.
30. **O. Kienzle**, F. Ernst, M. Rühle: *Quantitative HRTEM of the $\Sigma 3$ (111) Grain Boundary in Strontium Titanate*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin, 1996, CD-ROM.
31. **R. Schweinfest**, F. Ernst, G. Möbus, T. Wagner: *Structure of the Al/MgAl₂O₄ Interface Determined by Quantitative HRTEM*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin, 1996, CD-ROM.

32. **A. Recnik**, M. Ceh, F. Ernst: *Quantitative HRTEM Analysis of Antiphase Boundaries in SrO-Doped Calcium Titanate*, in: *Electron Microscopy 1994, vol. 2A*, Eds. B. Jouffrey and C. Coliex. Les Editions de Physique (Les Ulis), pp. 405–406, 1994.
33. F. Ernst, **D. Hofmann**: *Quantitative HRTEM of Grain Boundaries*, in: *Electron Microscopy 1994, vol. 2A*, Eds. B. Jouffrey and C. Coliex. Les Editions de Physique (Les Ulis), pp. 121–122, 1994.
34. S. K. Streiffer, S. Stemmer, W. Y. Hsu, F. Ernst, R. Raj, M. Rühle: *The Structure And Distribution of 90 Degrees Domains in PbTiO₃ Ferroelectric Thin Films*, in: *Fifty-second Annual Meeting, Microscopy Society of America and Twenty-Ninth Annual Meeting. Microbeam Analysis Society* (Eds. G. W. Bailey and A. J. Garratt-Reed), pp. 564–565, 1994..
35. **D. Hofmann**, G. Möbus, F. Ernst: *Quantitative HRTEM of Incoherent Twin Boundaries in Copper*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al.. Universidad de Granada (Granada), pp. 513–514, 1992.
36. G. Möbus, M. Hohenstein, F. Phillipp, F. Ernst, G. Necker, M. Rühle: *A Procedure to Measure the Chromatic Contrast Transfer Envelope of High-Resolution Electron Microscopes*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al... Universidad de Granada (Granada), pp. 531–532., 1992.
37. W. Mader, F. Ernst, G. Möbus: *HRTEM of Interfaces in Crystalline Solids*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al... Universidad de Granada (Granada), pp. 477–481, 1992.
38. S. Wen, F. Ernst, H. Jaeger, W. Mader P. Majewski, M. Marie, G. Petzow, M. Rühle: *HRTEM observation of the phase transition from Bi₂Sr₂CaCu₂O₈ to ₂Sr₂Ca₂Cu₃O₁₀*, in: „, Proceedings, annual meeting, Electron Microscopy Society of America. Cambridge University Press (CUP) (1990), pp. 94-95.

Publications as Editor

1. F. Ernst, M. Rühle: *High-Resolution Imaging and Spectrometry of Materials*. Springer Series in Materials Science, Springer Verlag (New York), 2002.
2. F. Ernst, J. Mayer, M. Rühle: *Journal of Microscopy* vol. 194, 1999.
3. W. Mader, F. Ernst: *Physica Status Solidi (a)* vol. 166, 1998.

Other Publications

Patents

1. P. Williams, S. Marx, F. Ernst, **A. Agaponova**: *Enhanced Activation of Self-Passivating Metals*. Patent, US 12371774B2, 2025-07-29.
2. Frank Ernst, Todd Johns, Ronald Edmonson, Cyprian Adair William Illing, Peter C. Williams, Christina Semkow: *Activation of Self-Passivating Metals Using Reagent Coatings for Low-Temperature Nitrocarburization in the Presence of Oxygen-Containing Gas*. Patent Application, filed April 28, 2022. Publication Number: 20220364216..
3. **Z. Ren**, F. Ernst: *Metal Carburization Process To Produce a Uniform, Concentrated Solid Solution of Interstitial Carbon Workpiece and Articles Made From Same*. Patent, US 11,066,735 B2.
4. **Zhe Ren**, Frank Ernst: *Metal Carburization Process to Produce a Uniform Concentrated Solid Solution of Interstitial Carbon Workpiece and Articles Made From Same*. US Patent Application, 56/654,367, filed on October 16, 2019..
5. **Zhen Li**, Frank Ernst, Harold Kahn, Arthur Heuer: *Alloy Surface Activation by Immersion in Aqueous Acid Solution*. Patent, US 10,450,658.
6. P. Williams, S. Marx, F. Ernst, **A. Agaponova**: *Enhanced Activation of Self-Passivating Metals*. Patent, US 2019/10,214,805.
7. **Z. Li**, F. Ernst, X. Zhan, A.H. Heuer: *Inducing Ferromagnetism in Ni-Base Superalloys*. Invention Disclosure, 2017.
8. **A. Agaponova**, P. Williams, S. Marx, F. Ernst: *Enhanced Activation of Self-Passivating Metals*. Patent Application, US 2016/0032442A1.
9. **Z. Li**, F. Ernst: *Alloy Surface Activation by Immersion in Aqueous Acid Solution*. PCT Application, 2015.
10. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. Japan Patent JP 5079963 B2, filed November 21, 2012.
11. **Chi-Hong Liao**, Uziel Landau, Frank Ernst: *A Non-Cyanide Alkaline Electroplating Bath Formulation for Copper Electrodeposition on Metals*. Provisional patent application, 2011.
12. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. US Patent application 10311716, filed February 7, 2003.

13. F. Ernst, G. M. Michal, A. H. Heuer: *Surface Hardening of Ti Alloys by Gas-Phase Nitridation: Kinetic Control of the Nitrogen Activity*. US Patent 2006/0048862 A1.
14. R. W. Bruce, A. H. Heuer, G. M. Michal, F. Ernst: *Titanium Treatment to Minimize Fretting*. US Patent 2006/0289088 A1.
15. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. Patent No. 01947428.7-2119-EP0107461, issued June 29, 2001.

Invited Presentations

1. *Alloy Surface Engineering by Interstitial Carbon or Nitrogen*. Colloquium, Texas A&M University. March 03, 2025.
2. *Alloy Surface Engineering by Infusion of Carbon or Nitrogen*. Colloquium, University of Alabama. November 09, 2023.
3. *Alloy Surface Engineering by Infusion of Interstitial Atoms*. Colloquium, Rice University. November 10, 2022.
4. *A New Perspective on Alloy Surface Engineering by Infusing High Concentrations of Interstitial Atoms*. Colloquium, University of Central Florida. March 12, 2021.
5. *Progress in Surface Engineering by Concentrated Interstitial Solute*. Colloquium, University of California at Davis. October 31, 2017.
6. *Materials Design by Surface Engineering*. Erich Bloch Symposium, University of Buffalo. June 1, 2017.
7. *Surface Engineering by Concentrated Interstitial Solute*. Colloquium, University of Maryland. May 5, 2017.
8. *Basic and Exploratory Research in the Department of Materials Science and Engineering at CWRU*. Army Research Laboratory, Aberdeen, MD, USA. February 14, 2017.
9. *The Department of Materials Science and Engineering*. Colloquium, Chemistry Department, College of Arts and Sciences, CWRU. October 28, 2016.
10. *Metals and Metalworking: DMSE as R&D Partner*. German Metal Working Delegation Meeting, Cleveland, OH, USA. July 12, 2016.
11. *Interdisciplinary Research*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 13, 2016.

12. *The Need to See Small: Structure-Property Relationships*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 12, 2016.
13. *CWRU and SCSAM*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
14. *R&D Interests at CWRU*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
15. *Computational Modeling of Concentration-Dependent Interstitial Diffusion*. Third Symposium on Surface Hardening of Corrosion-Resistant Alloys: Modeling and Structure, held at MS&T-2014, Pittsburg, PA. October 15, 2014.
16. *Low-Temperature Infusion of Interstitial Solutes: A New Concept for Surface Hardening of Steel*. International Symposium on New Trends in Steel Making and Steel Design, Institut für Neue Materialien, Saarbrücken, Germany. November 11, 2013.
17. *Alloy Surface Engineering by Gas-Phase Infusion of Interstitial Solutes*. Colloquium at the University of Birmingham, Birmingham, UK. August 02, 2013.
18. *Alloy Surface Engineering by Colossal Supersaturation with Interstitial Solutes*. Colloquium at the Christian-Albrechts-University, Department of Materials Science, Kiel, Germany. December 17, 2012.
19. *Surface Engineering of Alloys by Colossal Supersaturation with Interstitials*. Colloquium at the Institute of Engineering Innovation School of Engineering, The University of Tokyo, Tokyo, Japan. February 24, 2012.
20. *Colossal Supersaturation with Interstitial Solutes: Physical Principles and Potential for Alloy Surface Engineering*. WPI-AIMR Annual Workshop, Tohoku University, Sendai, Japan. February 2 to 23, 2012.
21. *Colossal Supersaturation: A Powerful Concept for Surface Engineering of Alloys*. ASM Young Member Night, University Club, Pittsburg, USA. February 16, 2012.
22. *Alloy Surface Hardening by Inward Diffusion of Interstitial Solutes*. Spring Seminar Series, Case School of Engineering, Cleveland, Ohio. March 25, 2011.
23. *Colossal Supersaturation with Interstitial Solutes: Perspectives for Surface Engineering of Medical Devices*. Materials Science & Technology 2010, Houston, Texas, USA. October 17 to 21, 2010.
24. *Thermodynamics of Low-Temperature Surface Carburizing*. 18th Congress of the IFHTSE, Rio de Janeiro, Brazil. July 26-30, 2010.

25. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. Kyoto University, Kyoto, Japan. June 27 to July 5, 2010.
26. *Carbide-Metal Interfaces in Low-Temperature-Carburized Structural Alloys*. XIII International Conference on Intergranular and Interface Boundaries (iib2010), Mie, Japan. June 27 to July 2, 2010.
27. *Strain State of Pt Catalyst Nanoparticles*. The 2nd International Symposium on Advanced Microscopy and Theoretical Calculations (AMTC2), Nagoya, Japan. June 24-26, 2010.
28. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. The Ohio State University, Columbus, Ohio. May 10, 2010.
29. *Applications Transmission Electron Microscopy in Materials Research*. CUNY - The City University of New York, New York, USA. April 29, 2010.
30. *Perspektiven höchstauflösender Mikrocharakterisierung in der Festkörper- und Materialforschung (Perspectives of High-Resolution Microcharacterization in Solid-State and Materials Research..* Institut für Festkörperforschung, Forschungszentrum Jülich, Germany. November 23, 2009.
31. *Surface Engineering of Medical Devices by Low-Temperature Carburization*. ASM Conference on Opportunities for Next Generation Implants. Cleveland. Ohio USA August 5-7, 2008
32. *Impact of Low-Temperature Carburization on the Microstructure of Structural Alloys*. 35th International Conference On Metallurgical Coatings And Thin Films (ICMCTF) 2008. San Diego, USA. April 28 to May 2, 2008
33. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Surface Engineering of Structural Alloys*. Colloquium at the University of Toledo. Toledo, USA. March 20, 2008
34. *Impact of Applied Electric Fields on Diffusion Reactions at Metal-Oxide Interfaces*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
35. *The Carbide-Austenite Interface in Austenitic Stainless Steels with a Colossal Supersaturation of Carbon*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
36. *Diffusion Reactions at Metal-Oxide Interfaces: The Effect of an Applied Electric Field*. 2006 TMS Annual Meeting & Exhibition, San Antonio, Texas, USA. March 12-16, 2006.
37. *The Microscopy and Microanalysis Facility at Case: Enabling Tools for Nanotechnology*. Functional Nanomaterials. Cleveland, Ohio. USA November 17, 2004

38. *TEM Studies of Cu₂O-Si and Cu₂O-InP Interfaces Made by Epitaxial Electrodeposition*. Microscopy & Microanalysis, Savannah, Georgia, USA. August 1-5, 2004.
39. *Nanoscale Characterization of Epitaxial Cu₂O Films Made by Electrodeposition*. Electronic Materials Conference (TMS), Notre Dame. June 23-25, 2004.
40. *Colossal Supersaturation: A New Concept for Surface Engineering*. The International Conference On Metallurgical Coatings And Thin Films ICMCTF, San Diego, California, USA. April 19-23, 2004.
41. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Advanced Structural Alloys*. Colloquium at the Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany. March 23, 2004.
42. *Microstructure and Phases of Cu-In-Se*. National CIS R & D Team Meeting, Orlando, Florida Solar Energy Center, Cocoa. November 13-14, 2003.
43. *Surface Hardening of Austenitic Stainless Steels by Colossal Supersaturation with Interstitial Solutes*. Structure-Property Relationships in High-Performance Ferrous-Based Systems Possessing Composite-Like Structures, Symposium at Materials Science & Technology 2003, Chicago. November 9-1, 2003.
44. *Alloy Surface Hardening by Colossal Supersaturation with Interstitial Solutes*. Colloquium of the Materials Science and Engineering Program, Rochester Institute of Technology. October 28, 2003.
45. *Colossal Supersaturation -A New Concept for Surface Engineering*. Colloquium of the Department of Materials Science and Engineering. The Ohio State University, Columbus. October 22, 2003.
46. *Liquid-Phase Deposition of α -Cu-In-Se Thin Films*. Space Photovoltaic Research and Technology Conference (SPRAT), Ohio Aerospace Institute. September 16-18, 2003.
47. *Characterization of Materials at the Nanometer Scale: Key to Nanotechnology*. NASA Tutorial on Nanotechnology, NASA Glenn Research Center, Cleveland, Ohio. August 23, 2003.
48. *Assessing Mechanical Properties of Interatomic Bonds Across Heterointerfaces*. Frontiers of Electron Microscopy in Materials Science (FEMMS-2003), Claremont, California. October 5-10, 2003.
49. *Structure of Cu₂O/Si Interfaces, Made by Epitaxial Electrodeposition*. Structure at the Limit of Resolution, Symposium at the 105th Annual Meeting of the American Ceramic Society, Nashville. April 27-30, 2003.

50. *Atomistic Structure and Composition of Grain Boundaries and Heterointerfaces*. Colloquium of the Materials Science and Engineering Department, University of Pittsburgh. March 20, 2003.
51. *Quantitative HRTEM and Modeling of Interfaces*. First Kyoto Workshop on Computational Materials Science, Kyoto (Japan). March 01-04, 2003..
52. *Advances in Assessing the Atomistic Structure of Grain Boundaries and Heterointerfaces*. Colloquium of the Engineering Research Institute, The University of Tokyo, Japan. February 28, 2003.
53. *Atomic Resolution and Bonding of Interfaces*. Colloquium of the Materials Engineering Department, University of Colorado in Boulder. January 28, 2003.
54. *High-Resolution Electron Microscopy*. Workshop on Diagnostics, Imaging and Spectroscopy, Cleveland, Ohio (USA). April 26, 2002.
55. *Assessing Interface Structures and Interfacial Atomic Bonds by Quantitative HRTEM*. Microscopy Society of Northern Ohio, Winter Meeting, Cleveland, Ohio (USA). February 13, 2002.
56. *Analysis of Interface Structure by Quantitative HRTEM*. Colloquium of the Department of Physics and Astronomy, Arizona State University, Tempe, Arizona (USA). December 04, 2001.
57. *Bridging the Gap from Atoms to Applications by Advanced Methods of Transmission Electron Microscopy*. United Engineering Foundation Meeting on "Integrating Materials Science into Engineering Structures and Devices," Lake Arrowhead, California (USA). November 11-15, 2001.
58. *Quantitative Analysis of Interface Structure by HRTEM*. Annual Meeting of the Microscopy Society of America, Long Beach, California (USA). August 05-09, 2001.
59. *Structure and Bonding of Metal/Oxide Interfaces*. Gordon Conference on High-Temperature Corrosion, New London, New Hampshire (USA). July 22-27, 2001.
60. *Atomistic Structure of Interfaces: Progress in Experimental Methods and Modeling*. Case School of Engineering, Faculty Seminar Series 2000-2001, CWRU, Cleveland (USA). May 1, 2001.
61. *Ge-Si Nanostructures for Quantum Effect Electronic Devices*. Microscopy of Semiconducting Materials, Oxford, UK. March 25-29, 2001.
62. *Progress in Materials Science by Synergy of High-Resolution, Analytical, and In-Situ TEM*. Workshop on Future Perspectives of Transmission Electron Microscopy, Stuttgart, Germany. January 17-18, 2001

63. *Self-Assembly of Ge-Si Dots by Coherency Strains*. DSRC/DARPA Workshop on Self-Assembly in Manufacturing, Strategic Analysis Conference Center, Arlington, Virginia. December 6-7, 2000.
64. *Progress in Determination of Defect Structures by Iterative Digital HRTEM Image Matching*. EUREM 12, Brno (Czech Republic). July 9-14, 2000.
65. *Grain Boundaries in Strontium Titanate*. Japan Fine Ceramics Center Workshop on Grain Boundaries in Ceramics. Nagoya (Japan), March 16-17, 2000.
66. *Structure, Composition, and Bonding of Grain Boundaries in Metals*. 37th Metallurgists' Day and 53rd Annual Technical Meeting of the Indian Institute of Metals. Kanpur (India) November 14-16, 1999.
67. *Atomistic Structure of Interfaces: Progress in Experimental Techniques and Modeling*. 3rd Slovenian-German Seminar, Bled, Slovenia. November 19-21, 1998.
68. *Ge-Si-Nanostructures for Quantum Effect Electronic Devices*. Technion-Israel Institute of Technology, Haifa, Israel. June 14, 1998.
69. *High Precision Assessment of Interface Structures by HRTEM*. 32nd Annual Meeting of the Israel Society for Microscopy, The Weizmann Institute of Science, Rehovot, Israel. June 11, 1998.
70. *Analysis of Interface Structures by Quantitative HRTEM*. MCEM 97, Portoroz, Slovenia. October 5-8, 1997.
71. *Analysis of Interface Structures by Quantitative HRTEM*. EMAG 97, Cambridge, Great Britain. September 2-5, 1997.
72. *Solving Interface Structures by Quantitative High-Resolution Electron Microscopy*. Physical Colloquium, University of Vienna, Austria. December 11, 1997.
73. *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Fall Meeting of the Materials Research Society, Boston. December 1-5, 1996.
74. *Quantitative HRTEM of Interfaces and Structure of Metal/Ceramic Interfaces*. Micromat 1996. Rio de Janeiro, October 13-16, 1996.
75. *Analysis of Grain Boundary Structures by Quantitative HRTEM*. Annual Meeting of the Microscopy Society of America (MSA), Minneapolis. August 11-15, 1996.
76. *Interface Dislocations*. International Workshop on Plastic Deformation of Structural Materials at High Temperatures, Schloss Ringberg, Tegernsee, Germany. July 14-17, 1996.

77. *Quantitative High-Resolution Electron Microscopy of Interfaces*. 7th International Conference on Intergranular and Interphase Boundaries in Materials (iib95), Lissabon, Portugal. June 26-29, 1995.
78. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Modeling*. Colloquium of the Institut für Physikalische Chemie und Elektrochemie, University of Hannover, Germany. May 8, 1995.
79. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Simulation*. 13th International Congress on Electron Microscopy, Paris, France. July 18-22, 1994.
80. *Quantitative HRTEM of Twin Boundaries*. Annual Congress of the German Physical Society, Münster, Germany. March 21-25, 1994.
81. *Quantitative High-Resolution Electron Microscopy as a Method to Determine the Structure of Grain Boundaries*. German Physical Society, Spring Meeting, Münster. March 21-25, 1994.
82. *Lattice Mismatch Accommodation by Misfit Partial Dislocations*. Annual Meeting of The Materials Society (TMS), San Francisco. February 27-March 3, 1994.
83. *Quantitative HRTEM of Grain Boundaries*. International Symposium on High-Voltage and High-Resolution Electron Microscopy, Stuttgart, Germany. February 21-24, 1994.
84. *Assessment of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Workshop on Microcharacterization of Materials and Devices, Göttingen, Germany. January 6-7, 1994.
85. *Quantitative Evaluation of HRTEM Images*. Multinational Congress of the Austrian, German, and Swiss Electron Microscopy Societies, Zürich. Switzerland, September 1993..
86. *HREM of Metal/Ceramic Interfaces*. Annual Meeting of the French Society of Electron Microscopy (SFME), Lyon, France. June 29-July 4, 1993..
87. *Misfit Dislocation Networks in 111-GeSi/Si-Interfaces*. Colloquium of the IV. Physikalisches Institut, University of Göttingen. June 24, 1993.
88. *Relaxation of Misfit Stresses in Epitaxial Layers*. Physical-Chemical Colloquium, University of Stuttgart, Germany. June 22, 1993.
89. *Phase Transformations at Metal/Ceramics Interfaces*. International Workshop on Reactive Phase Formation at Interfaces and Diffusion Processes, Aussois, France. May 21-2, 1993..

90. *Misfit Dislocation Structure of GeSi/Si-{111}-Interfaces*. European Science Foundation Conference on Extended Defects in Semiconductors, Holzgau, Germany. August 29-September 3, 1992.
91. *Quantitative Analysis of Intensities in HRTEM Lattice Images*. Annual Congress of the German Electron Microscopy Society (DGE), Darmstadt, Germany. September 8-13, 1991.
92. *Misfit Dislocation Networks in GeSi/Si Interfaces Grown by Liquid Phase Epitaxy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. February 11, 1991.
93. *HRTEM of the Cu/MnO Interface in an Internally Oxidized CuMn Alloy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. April 23, 1990.
94. *HRTEM of Metal/Ceramic Interfaces in Internally Oxidized Alloys*. 4th Workshop on Characterization and Properties of Interfaces in Crystals, Delft, Netherlands. June 5-9, 1989.
95. *High-Resolution Electron Microscopy of Semiconductor Heterointerfaces*. Colloquium of the Max-Planck-Institut für Festkörperforschung, Stuttgart. Germany. May 8, 1989.
96. *High-Resolution Electron Microscopy of Cu/Al₂O₃ Interphase Interfaces*. Colloquium of the Institut für Metallphysik, University of Göttingen, Germany. April 18, 1989.

Courses Taught and Developed*

Univ	Code	Course
Stuttgart		Kristallografie und Beugung* (Crystallography and Diffraction)
Stuttgart		Werkstoff-Analytik 1* (Analytical Methods of Materials Science 1*)
Stuttgart		Halbleiter Werkstoffe der Mikorelektronik und Photovoltaik* (Semiconductor Materials for Microelectronics and Photovoltaics*)
CWRU	EMSE-102	Introduction to Materials Science and Engineering*
CWRU	EMSE-228	Mathematical and Computational Methods for Materials Science and Engineering*
CWRU	EMSE-290	Materials Laboratory III
CWRU	EMSE-314	Electrical, Optical, and Magnetic Properties of Materials*
CWRU	EMSE-327	Thermodynamic Stability and Rate Processes*
CWRU	EMSE-368/468	Scientific Writing*
CWRU	EMSE-398-399	Senior Project in Materials Science and Engineering
CWRU	EMSE-465	Surface Engineering of Materials*
CWRU	EMSE-509	Conventional Transmission Electron Microscopy*
CWRU	EMSE-512	Advanced Methods of Transmission Electron Microscopy*
CWRU	EMSE-515	Analytical Methods of Materials Science and Engineering*

Consulting

Period	Client	Activity
Since 07/2026	Cooley LLP	Expert Witness for Tower in: Global-Foundries vs Tower Semiconductor
08/2025- 01/2026	Cooley LLP	Expert Witness for Caihong in: Corning Incorporated vs Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd.
03/2025- 07/2025	Merchant & Gould P. C.	Expert for Caihong in: Inter Partes Review Petitions of Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd. vs Corning Incorporated
Since 02/2026	Bosch	Consultant for Semiconductor Device Processing
02/2025- 12/2025	Bosch	Consultant for Semiconductor Device Processing
07/2018- 01/2024	Reichman & Jorgensen	Expert Witness for Arconic in: Arconic Corporation et al vs Novelis Inc. et al
05/2018- 08/2018	Voss Aerospace In- dustries Inc.	Mechanical Properties of Aluminum Alloys
01/2017- 12/2024	KFAS	Proposal Reviewer for Kuwait Foundation for the Advancement of Sciences
03/2016- 04/2018	Atotech	Development of a Know-How Management System
03/2014- 08/2021	Skolkovo	Proposal ("Expert") Reviewer
04/2014- 06/2014	Voss Aerospace In- dustries Inc.	Microstructure of Precipitation-Hardened Aluminum Alloys

07/2012	Powdermet	Microscopic Structure of Novel Powders
07/2011– 06/2012	Atotech	Microstructure of Electroplated Cu for Microelectronic Devices
04/2011– 10/2011	AlSher Titania LLC	Structure and Composition of Titania Nanoparticles
01/2011– 12/2017	Pocared	Microstructure of Coatings for Pharmaceutical Containers
09/2008	TRW Automotive Engine Components	Nanohardness and Auger Electron Spectroscopy Studies of Valves
04/2008	DoD	Proposal reviewer for SERDP program
04/2007	DoD	Proposal reviewer for SERDP program
10/2006– 05/2007	Bird Technologies Group	Scanning Electron Microscopy and Analysis of Resistors
11/2004– 04/2005	Patent Litigation	Expert Witness Assistance for Experimental Studies of Coatings
05/2004– 11/2004	OMG Americas	Transmission Electron Microscopy Analysis of Novel Nanoparticles
12/2003– 06/2005	Moen Incorporated, North Olmsted, Ohio	Transmission Electron Microscopy Studies of Coatings
06/2003– 12/2003	IMQ Corporation, Chesterland, Ohio	Transmission Electron Microscopy Studies on the Effect of Boron on the Surface Structure of Ferritic Steels.
02/2003– 05/2003	Inst. of Semiconductor Technology, Stuttgart, Germany	Transmission Electron Microscopy Analysis of Defect Structures in Ultra-thin SiGe Buffer Layers

EXHIBIT B

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Employment History

RF DESIGN CONCEPTS, LTD., EXPERT WITNESS, RF DESIGN, TURN KEY SOLUTIONS 02/2012–Present

Principal & Owner

- Subject Matter Expert in RF/Microwave (Wireless) Hardware: RF Power Amplifiers/Transmitters, RF Front-end Modules/Transceivers, Power Management, Electromagnetics (Signal integrity, antennas, filters, etc.), Wireless Charging, Semiconductors, Digital Electronics, Device Packaging
 - Consultant, Satcom SSPA, C-band RFPA– Trak Microwave
 - Consultant, 24 GHz Radar Front-End – The Ohio State University
 - Consultant, Ultra compact high efficiency load adaptable RFPA– Draper Labs
 - Consultant, High voltage MEMS packaging – Bright Silicon (Aspen Microsystems)
 - Consultant, Ultra high efficiency wireless charging– NuCurrent, Inc.
 - Consultant, Ultra high efficiency RF plasma generation– Advanced Energy
 - Consultant, DARPA MIS superconductor fabrication program– Grid Logic, Corp.
 - Consultant, Quantum devices– ColdQuanta, Inc.
 - Consultant, LTCC “RF” module for optical communications– AvoPhotonics, Inc.
 - Consultant, RF front-end module– Trellisware
 - Consultant, Market/System ecology analysis of nascent technologies– Invensas
 - Product Manager (Consultant)– UX Lab, Samsung Electronics
 - Custom Design of RF power amplifier solutions for commercial wireless (cdma, WCDMA, LTE), defense (radar, EW, SIGINT, etc., medical (MRI, RF ablation), and industrial (RF induction heating, metal sputtering, plasma generation, etc.)
 - Custom Design of DC power management systems, including Switched-mode (Buck, Boost, Buck-boost, etc.) and linear (LDO) power supply designs
- Expert Witness, Patent Litigation & Technical Due Diligence– Fish & Richardson, Wilmer Hale, Kirkland & Ellis, Perkins Coie, Slayden Grubert Beard, Pillsbury Winthrop Shaw Pittman, O’Melveny, Finnegan, Wolf Greenfield, Shelton Coburn, Alston & Bird, Weil, Hoynig Rokh Monegier, Susman Godfrey

MERCURY SYSTEMS, INC., RF POWER AMPLIFIER BUSINESS UNIT

08/2013–03/2014

Technical Director

- Responsible for all aspects of Mercury Systems’ RF Power Amplifier Business Unit
 - Full P&L responsibility for \$2.5M RF Power Amplifier Business Unit
 - Responsible for engineering, sales, manufacturing, and test

- Develop new business opportunities through direct customer contact and proposals (Technical concept, cost models, etc.)
- Lead and manage new product development and sustaining engineering efforts
- Manage development and production schedules, material planning, product testing
- Manage customer relationships

LGS BELL LABS INNOVATIONS, R&D HARDWARE ENGINEERING

05/2011–12/2012

Sr. R&D RF Hardware Engineer (Technical Ladder, Grade 315)

- Development of high power, highly linear and efficient RF transmitter/transceiver solutions supporting major wireless standards (GSM/EDGE, cdma, WCDMA, LTE)
 - First-pass design success of (200W P1dB) 850MHz and (250W P1dB) 900MHz band Doherty RF transmitters
 - First-pass design success of highly linear, thermally compensated TX lineups for 746-960MHz and 1805-2170MHz broadband transmitters
 - Development of high power (46dBm avg), broadband, highly linear RF transmitter
- Technical leadership/project management
 - Coordinate activities of multi-disciplinary teams of RF/electrical engineers, mechanical engineers, drafters, and technicians
 - Insure adherence to proper engineering design practices and manufactureability; Internal design review

AETHERCOMM, INC., ADVANCED DEVELOPMENT GROUP

07/2008–02/2011

Technical Manager/Principal Engineer

- Design Engineering Responsibilities
 - Responsible for complete execution of rapid product development of pulsed/CW, wideband, high power RF transmitters for radar and communications on extremely accelerated schedules per MIL-STD's: Silicon LDMOS, GaAs FET & HBT, and GaN HEMT device technologies
 - Customer specification to top-level design, complete electrical design (RF link budget analysis, RF circuitry, DC power management, control logic, telemetry, fault protection, etc.), bill of materials (BoM), test procedures, and design verification testing
 - Lead, manage, and coordinate activities of multi-disciplinary teams of RF/electrical engineers, mechanical engineers, drafters, and technicians
 - Identify and execute, internal R&D programs, including SBIR's
- Technical Management Responsibilities
 - "Cradle to grave" management of all technical aspects of product development to insure successful, on time delivery of custom RF products
 - Product documentation and design milestone reviews: Compliance matrix, ECN's, test plans (ATP, DVT, TDR), final product approval, and post mortem reports
 - Maintain competent & effective staff through recruitment, mentoring, and performance planning of technical personnel
 - Insure adherence to good engineering design practice and manufactureability; Review designs through all engineering phases

- Support upper management and marketing through customer interaction, speaking in technical forums, and writing technical publications
- Resource & budget planning, allocation, and scheduling of all group activities

SAMSUNG ELECTRONICS, TECHNOLOGY INTELLIGENCE & COLLABORATION

03/2007–07/2008

Section Manager, Technology Strategy Team

- Technical feasibility and business opportunity analysis of bottom-up nanotechnology for next generation digital storage, wearable electronics, and non-lithographic semiconductor technologies
- Market and technology analysis of (low-power wireless) biomedical electronics
- Market and technology analysis of THz imaging for security, healthcare, and industrial applications

RF MICRO DEVICES, INFRASTRUCTURE PRODUCTS GROUP

07/2006–02/2007

Group Leader/Staff Engineer

- Advanced Product Development: GaN HEMT dual carrier UMTS (2.14 GHz WCDMA) Doherty basestation transmitter (240 W max./40 W avg. RF power)
- Led development of scalable GaN HEMT non-linear simulator model
- Project planning, product documentation (PRD, PPA, etc.), and test plan development

NOKIA CORPORATION, NOKIA RESEARCH CENTER

04/2001–03/2006

Senior Research Engineer, Radio Technologies Laboratory

- Advanced RF Research:
 - *RF Circuits*: Complete design, build, test, and system integration of novel LDMOS based Chireix Outphasing (basestation) transmitter for UMTS systems (120W max./20W avg. RF power)
 - *RF Systems*: Development of OFDM RF system simulator (IEEE 802.11a WLAN & ETSI EN-300 744 DVB-T PHY); Baseband PAPR reduction techniques for OFDM radio systems
 - *Power Management*: Complete design, build, and test of high power, high precision, 16 bit digitally-controlled switched-mode power supply
 - *Electromagnetics*: Compact antenna and (metamaterial) filter design
- Project Management:
 - Technical Liaison: Advise on new technologies offered by established and new venture organizations in collaboration with Nokia Ventures and Nokia Mobile Phones New Technology Sourcing
 - Advise and steer corporate funded university research projects
 - Group leader, *Leading Science 8* program, led market research and analysis for “Network and Infrastructure Solutions in Low Average Revenue per User (ARPU) Markets”

WESTERN DIGITAL CORPORATION/ADAPTEC INC.

01/1995–08/1996

Digital Design Engineer, ASIC Design

- Timing Analysis: Static timing analysis of ASIC critical path delays; Custom cell characterization and library generation
- Clock Tree & Pad Cell Drivers: Clock tree timing analysis and design; Custom ASIC pad cell design; Drivers for memory cells
- Power ON Logic: Developed “No re-spin” solution for faulty ASIC ACK/Power Enable
- Meta-stability Analysis: Circuit and system level analysis of meta-stable logic

Academic History

UNIVERSITY OF COLORADO, BOULDER

03/2013–03/2014; 08/2015–08/2016

Adjunct Professor, Department of Electrical, Computer, & Energy Engineering

UNIVERSITY OF CALIFORNIA, SAN DIEGO

08/2004–08/2008

Visiting Scholar (Part-time), Department of Electrical & Computer Engineering

- Established LoA between UC San Diego, Cree Microwave, Inc., and Nokia for collaborative research on the Envelope Tracking (ET) highly linear and efficient transmitter architecture
- Coordination of corporate sponsored research on linear, highly efficient transmitter architectures

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

09/1996–03/2001

Doctor of Philosophy, Electrical & Computer Engineering– Solid-State Electronics (major), Analog/Digital Circuits (minor); Dissertation title: High Efficiency Switched-Mode Power Amplifiers for Wireless Communications

CALIFORNIA STATE UNIVERSITY, FULLERTON

08/1994–01/1996

Master of Science, Electrical & Computer Engineering– Analog/Digital Circuits (major)

UNIVERSITY OF CALIFORNIA, BERKELEY

08/1990–05/1993

Bachelor of Science, Physics

Selected Publications

T. Hung, D. K. Choi, L. E. Larson, P. M. Asbeck, “CMOS Outphasing Class-D Amplifier With Chireix Combiner,” *IEEE Microwave and Wireless Components Letters*, 8:619-621, August, 2007.

I. Hakala, D. K. Choi, L. Gharavi, N. Kajakine, J. Koskela, R. Kaunisto, “A 2.14-GHz Chireix Outphasing Transmitter,” *IEEE MTT Trans*, 4:2129-2138, June, 2005.

D. Qiao, D. Choi, Y. Zhao, D. Kelly, T. Hung, D. Kimball, M. Li, P. Asbeck, “Antenna Impedance Mismatch Measurement and Correction for CDMA Transceivers,” *IEEE MTT-S Digest*, WE3C-4, Long Beach, CA, June, 2005.

D. K. Choi and S. I. Long, “The Effect of Transistor Feedback Capacitance in Class-E Power Amplifiers,” *IEEE Trans. on Circuits and Systems I*, Vol. 50, Issue 12, pp. 1556-1559, December, 2003.

D. K. Choi and S. I. Long, "Finite DC Feed Inductors in Class E Power Amplifiers– A Simplified Approach," *IEEE MTT-S Digest*, TH2D-5, Seattle, WA, June, 2002.

D. K. Choi and S. I. Long, "A Physically Based Analytic Model of FET Class E Power Amplifiers– Designing for Maximum PAE," *IEEE MTT Trans*, 47:1712-1720, September, 1999.

Other Professional/Personal Information

US citizen, US Secret & Top Secret/SCI (In-active) clearance, Senior member IEEE, California State MICRO Fellow (1996), fluent in the Korean language, conversant in the French language.

EXHIBIT C

Jeffrey Bokor, Ph.D.

Paul R. Gray Distinguished Professor Emeritus
EECS Department
University of California
Berkeley, CA 94720

Phone: (510)642-4134
jbokor@berkeley.edu
www.eecs.berkeley.edu/~jbokor

Expertise

- Electronics
- Nanofabrication
- Nanotechnology
- Device Physics
- Nanomagnetism and Spintronics
- Optics and Optoelectronics
- Laser Technology
- Semiconductor Physics
- Semiconductor Process
- VLSI Technology

Professional Summary

Education

<u>Year</u>	<u>College/University</u>	<u>Degree</u>
1980	Stanford University	Ph.D., Electrical Engineering
1976	Stanford University	MS, Electrical Engineering
1975	Massachusetts Institute of Technology	BS, Electrical Engineering

Employment History

From:	1993	University of California, Department of Electrical Engineering and Computer Science, Berkeley CA
To:	2024	Berkeley, CA
Position:		
	2024-present	Emeritus Professor
	1993-2024	Professor
		▪ Department Chair, Department of Electrical Engineering and Computer Sciences, 2019-2021 ▪ Chair, EE Division, Associate Department Chair, Department of Electrical Engineering and Computer Science, 2017-2019 ▪ Associate Dean for Research, College of Engineering, 2012-2017 ▪ Joint Appointment: 2008-2024 , Faculty Senior Scientist, Lawrence Berkeley National Laboratory ▪ Joint Appointment: 2004-2012, Deputy Director for Science, The Molecular Foundry, Lawrence Berkeley National Laboratory ▪ Joint Appointment: 1993-2004, Group Leader for Advanced Lithography, Center for X-ray Optics, Lawrence Berkeley National Laboratory

Curriculum Vitae - Jeffrey Bokor

Current research interests:

- Nanoelectronic devices
- Spintronics
- Carbon electronics
- Ultrafast laser studies of materials

Classroom teaching:

- Undergraduate and graduate courses in semiconductor physics and technology
- Undergraduate and graduate courses in optical physics and engineering
- Freshman seminar introduction to electrical engineering

From:	1980	AT&T, Bell Laboratories
To:	1993	Holmdel and Murray Hill, NJ
Position:	1990-93	Ultra-Large Scale Integrated Circuits (ULSI) Technology Research Department. Department Head. Murray Hill, NJ Leader of group of 15 Ph.D. scientists, 35 total technical personnel. Responsible for directing research to create technologies relevant to silicon IC manufacturing in the year 2000. Also responsible for continuing improvement and operation of 20,000 sq. ft. Class 100 research cleanroom facility. Experience included: ▪ ULSI Process integration ▪ Deep sub-micron silicon MOSFET device physics ▪ Device and circuit design ▪ Advanced process and device modeling software ▪ Advanced lithography
	1987-90	Laser Science Research Department. Department Head, Holmdel, NJ. Leader of group of 10 Ph.D. scientists, 20 total technical personnel. Managed diverse program of basic research in optical physics. Conducted high visibility research program, including: ▪ X-ray lithography ▪ Nonlinear optics ▪ Picosecond optoelectronics ▪ Femtosecond phenomena in electronic materials ▪ Fiber optics ▪ Atomic physics
	1980-87	Quantum Electronics Research Department. Member of Technical Staff, Holmdel, NJ. Independently directed basic research, including: ▪ Nonlinear optics ▪ Picosecond optoelectronics ▪ Semiconductor device physics ▪ Ultraviolet laser technology

Curriculum Vitae - Jeffrey Bokor

Major Research Accomplishments (1980-Present)

- Demonstration of ultrafast (<10 psec) spin-orbit torque switching (2023)
- Demonstration of ultralow contact resistance between semimetal and monolayer semiconductors (2021)
- Demonstration of double-gate graphene nanoribbon transistors (2021)
- Demonstration of transfer-free synthesis of atomically precise graphene nanoribbons on insulating substrates (2021)
- Demonstration of spin-transfer-torque magnetic recording (2019)
- Demonstration of 7 nm magnetic memory using scanned probe magnetic tunnel junction readout (2018)
- Experimental demonstration of ultrafast (<10 psec) magnetic switching by pure charge current pulses (2017)
- High performance bottom-up chemically synthesized graphene nanoribbon transistors (2017)
- Experimental demonstration of the ultimate lowest switching energy set by fundamental physics (Landauer's principle) in single-bit operations on nanomagnetic memory bits (2016)
- Demonstration of spin transfer torque switching in sub-10 nm size magnetic tunnel junctions (2016)
- Demonstration of fluid-suspended magnetic particle manipulation using multiferroic magnetic actuation (2015)
- Direct optical detection of current induced spin accumulation in non-magnetic metals (2015)
- Demonstration of sub-nsec signal propagation in nanomagnetic logic (2015)
- Electric field control of magnetic domain walls for manipulation of magnetic particles in microfluidics (2014)
- High performance nanoscale solution processed carbon nanotube transistors (2013)
- Bottom-up chemically synthesized graphene nanoribbon transistors (2013)
- Nanoscale hyperspectral imaging with tapered plasmonic "Campanile" tip (2012)
- Demonstration of nanomagnetic logic signal propagation (2012)
- Demonstration of fundamental thermodynamic limit of energy dissipation in magnetic memory and logic (2011)
- Demonstration of bandgap and sub-band formation in graphene nanomesh superlattice transistors (2010)
- Self-assembly of gold nanoparticle chain structures for nanophotonics using DNA origami (2010)
- Measurement of Schottky contact junction capacitance, doping profiles and carrier mobility in single semiconductor nanowire and carbon nanotube transistors using ultrasensitive C-V measurements (2009)
- Label-free DNA biosensors using functionalized carbon nanotube transistors (2009)
- Directed assembly of nanophotonic device structure using DNA nanotechnology (2009)
- Theory of spin-dependent scattering in silicon MOSFET transistors (2009)
- Proposed new concepts for nanomagnetic logic technology (2008)
- Study of spin-dependent scattering from impurity donors in silicon MOSFETs for use in quantum computing (2007)

Curriculum Vitae - Jeffrey Bokor

- Measurement of spin coherence of antimony donors in silicon near silicon dioxide interfaces (2006)
- First observation of Stark tuning of donor electron spins in silicon, a key step towards implementation of quantum logic in silicon technology (2006)
- First monolithic integrated circuit with both carbon nanotube and silicon MOS transistors (2004)
- Development of a monolithic inkjet printhead array capable of sub-3 μm droplet generation (2004)
- Study of surface and bulk acoustic phonon excitations in superlattices using picosecond ultrasonics (2003)
- Invention and demonstration of “spacer lithography”, a sub-10 nm patterning technology for nano-catalysts and DNA label-free hybridization detection (2003)
- Development of 10 nm gate-length MOSFET devices (2002)
- Demonstration of extreme ultraviolet lensless Fourier transform holography (2001)
- Demonstration of 30 nm lithography using extreme ultraviolet radiation (2001)
- Demonstration of aberration testing and alignment optimization of a commercial prototype four-mirror ring field extreme ultraviolet optical system using at-wavelength interferometry (2000)
- Development of complementary silicide source/drain technology for sub-20 nm MOSFET devices (2000)
- Development of picosecond ultrasonic characterization of Mo/Si extreme ultraviolet multilayer reflectors (1999)
- Development of 17 nm gate-length FinFET devices (1999)
- First demonstration of the “FinFET”, a manufacturable double-gate MOSFET with 25 nm gate-length (1998)
- Development of at-wavelength defect detection metrology for extreme ultraviolet lithography (1998)
- Co-invention and demonstration of the phase-shifting, point diffraction interferometer, and application to high-accuracy wavefront metrology of extreme ultraviolet lithography optics (1996)
- First demonstration of working MOS devices fabricated with extreme ultraviolet lithography (1996)
- Direct, quantitative measurement of saturation velocity and overshoot velocity in MOS inversion layers (1994)
- Demonstration of interferometry at extreme ultraviolet wavelengths for characterization of lithography optics (1994)
- First direct observation of non-thermal energy distributions and thermalization dynamics in laser-excited metal films (1992)
- Development of 89 GHz f T Si MOSFET transistors with 0.15 μm channel length (1992)
- First feasibility demonstration of diffraction-limited extreme ultraviolet projection lithography (1990)
- First direct observation of surface state kinetics during semiconductor surface recombination (1989)
- Invention and demonstration of the coplanar vacuum photodiode, an ultrafast, high sensitivity photodetector for the visible and ultraviolet (1987)

Curriculum Vitae - Jeffrey Bokor

- First demonstration of time-resolved photoemission for measurement of picosecond electronic signals in integrated circuits (1986)
- First direct detection of band-gap states at GaAs Schottky barrier interfaces (1986)
- Application of time-resolved photoemission to the first measurement of picosecond electron dynamics on InP, GaAs, and Si surfaces (1985)
- Invention and demonstration of picosecond time-resolved ultraviolet photoemission spectroscopy for the study of electron dynamics at semiconductor surfaces (1985)
- Development of a source of 35.5 nm coherent radiation by seventh harmonic generation in a pulsed jet. At the time, this was the shortest wavelength source of coherent radiation yet demonstrated (1983)
- Invention of the pulsed jet harmonic generator for the production of extreme ultraviolet coherent radiation (1983)
- First picosecond pulse krypton-fluoride excimer laser (1982)
- First demonstration of an autoionization pumped laser (1982)
- First third harmonic generation using an excimer laser (1980)
- Development of the first tunable, single-mode excimer laser system (1980)

Technical Consulting

From: 2001 Rodel, Inc.
To: 2002 Newark, DE
Position: Technical advisory board member for extreme ultraviolet lithography

From: 2002 Alpha Golf, Inc.
To: 2002 Kensington, CA
Position: Technical consultant

From: 2002 University Technology Ventures, Inc.
To: present Pleasanton, CA
Position: Technical consultant

From: 2002 Veeco Metrology Group
To: 2006 Santa Barbara, CA
Position: Technical advisory board member

Litigation Support

Consulting and expert witness on patent validity and infringement in 30+ cases since 1998. Equipment inspections. Testimonial experience at deposition and trial. Full list available on request.

Curriculum Vitae - Jeffrey Bokor

Patents

1. Yang Yang, Jon Gorchon, Richard Wilson, Charles-Henri Lambert, Sayeef Salahuddin, Jeffrey Bokor, "Writing of a magnetic memory with electric pulses," US 10,388,349 B2, August 2019.
2. Thomas Schenkel, Cheuk Chi Lo, Christoph Weis, Stephen Lyon, Alexei Tyryshkin, Jeffrey Bokor, "Scalable quantum computer architecture with coupled donor-quantum dot qubits," US 8,816,325 B2, August 2014
3. Brian Lambson, Zheng Gu, David Carlton, Jeffrey Bokor, "Concave nanomagnets with widely tunable anisotropy," US 8,766,754 B2, July 2014
4. Ji Zhu, Jeff Grunes, Yang-Kyu Choi, Jeffrey Bokor, Gabor Somorjai, "Methods for Fabrication of Positional and Compositionally Controlled Nanostructures on Substrate," US 8,486,287 B2, July 2013
5. Jeffrey Bokor, Nathan C. Emley, David Carlton, "Nanomagnetic signal propagation and logic gates," US 8,134,441 B1, March, 2012
6. David Carlton, Nathan C. Emley, Jeffrey Bokor, "Nanomagnetic Register," US 8,138,874 B1, March, 2012
7. Jeffrey Bokor, Patrick Naulleau, "System For Interferometric Distortion Measurements That Define An Optical Path," US 6,559,952 B1, May, 2003
8. Jeffrey Bokor, Yun Lin, "Method And Apparatus For Inspecting Reflection Masks For Defects," US 6,555,828 B1, April, 2003
9. Jeffrey Bokor, Seongtae Jeong, "Multi-Level Scanning Method for Defect Inspection," US 6,484,306 B1, November 2002
10. Chenming Hu, Tsu-Jae King, Vivek Subramanian, Leland Chang, Xuejue Huang, Yang-Kyu Choi, Jakub Tadeusz Kedzierski, Nick Lindert, Jeffrey Bokor, Wen-Chin Lee, "FINFET Transistor Structures having a Double Gate Channel Extending Vertically From a Substrate and Methods of Manufacture," US 6,413,802 B1, July 2002
11. Jeffrey Bokor, Anthony M. Johnson, "High Speed Photodetector Having Dual Transmission Line Geometry," US 4,933,542 A, June 1990
12. Jeffrey Bokor, Anthony M. Johnson, Ralph H. Storz, "High Speed Circuit Measurements Using Photoemission Sampling," US 4,721,910 A, January, 1988

Professional Associations, Service, and Awards

- Fellow, Optical Society of America
- Fellow, American Physical Society
- Fellow, Institute of Electronics and Electrical Engineers
- Senior Scientist, Lawrence Berkeley National Laboratory
- American Association for the Advancement of Science
- Paul R. Gray Distinguished Professorship in Electrical Engineering and Computer Sciences, 2015-2024
- National Semiconductor Distinguished Professorship in Engineering, University of California, 2007-2015
- Deputy Director, Center for Energy Efficient Electronic Science, an NSF Science and Technology Center headquartered at Berkeley, 2010-2020
- Co-Chair, 5th Berkeley Symposium on Energy Efficient Electronic Systems, 2017

Curriculum Vitae - Jeffrey Bokor

- Co-Chair, 4th Berkeley Symposium on Energy Efficient Electronic Systems, 2015
- Co-Chair, 3rd Berkeley Symposium on Energy Efficient Electronic Systems, 2013
- Co-Chair, 2nd Berkeley Symposium on Energy Efficient Electronic Systems, 2011
- Co-Chair, 1st Berkeley Symposium on Energy Efficient Electronic Systems, 2009
- Scientific Advisory Board, Catalan Institute for Nanoscience (ICN), Autonomous University of Barcelona, 2007-2022
- Scientific Advisory Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 2006-2010
- IEEE Paul Rappaport Award for best paper of the year published in Electron Device Society journals, 2002
- DARPA Most Significant Technical Accomplishment Award, 2000
- Program Committee, CMOS devices, International Electron Devices Meeting, 1999-2000
- Scientific Advisory Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 1998-2002
- Technical Advisory Committee, Laser Science and Technology Program, Lawrence Livermore National Laboratory, 1996-1997
- General Co-chair, Quantum Electronics and Laser Science Conference, Baltimore, MD, 1997
- Program Committee, Topical Meeting on Extreme Ultraviolet Lithography, Boston, MA, 1996
- Users Executive Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 1994-1998
- Program Co-chair, Quantum Electronics and Laser Science Conference, Baltimore, MD, 1995
- Technical Council, Optical Society of America, 1991-1994
- Chairman, Technical Group on X-ray and XUV Physics, Optical Society of America, 1991-1994
- Program Committee, Topical Meeting on Soft X-ray Projection Lithography, Monterey, CA, April, 1992
- Cochairman, Topical Meeting on Soft X-ray Projection Lithography, Monterey, CA, April, 1991
- Cochairman, Topical Meeting on Short Wavelength Coherent Radiation, Generation and Applications, Monterey, CA, March, 1986
- Program Committee, Conference on Lasers and Electro-Optics (CLEO) '84, Anaheim, CA, June, 1984
- Program Committee, Topical Meeting on Laser Techniques in the Extreme Ultraviolet, Boulder, CO, March, 1984
- Program Committee, Topical Meeting on Excimer Lasers, Lake Tahoe, NV, January, 1983
- Program Committee, Topical Meeting on Laser Techniques for Extreme Ultraviolet Spectroscopy, Boulder, CO, March 1982

Curriculum Vitae - Jeffrey Bokor

Publications

1. D. Polley and J. Bokor, "A perspective on ultrafast magnetization switching in ferromagnets using helicity-independent optical pulses," *Journal of Physics D: Applied Physics*, vol. 58, no. 10, p. 102001, (2025).
2. X. Li, H. Singh, J. Lin, S. Zhang, B. Yi, J. Chatterjee, Z. Xiao, S. Mondal, N. Tamura, R. N. Candler, L. You, J. Bokor, and J. Hong, "An energy efficient way for quantitative magnetization switching," *npj Spintronics*, vol. 2, 2024-07-17 2024.
3. M. Goiriena, Z. Xiao, R. Steinhardt, V. Estrada, N. Tamura, R. V. Chopdekar, A. T. N'Diaye, A. Sepúlveda, G. G. Schlom, R. N. Candler, and J. Bokor, "Imaging of voltage-controlled switching of magnetization in highly magnetostrictive epitaxial Fe–Ga microstructures." *Nanoscale* **16**(18): 9021–9028 (2024).
4. D. Polley, A. Pattabi, A. Rastogi, K. Jhuria, E. Diaz, H. Singh, A. Lemaitre, M. Hehn, J. Gorchon, and J. Bokor, "Picosecond spin-orbit torque–induced coherent magnetization switching in a ferromagnet," *Science Advances*, vol. 9, p. eadh5562, 2023.
5. D. Polley, J. Chatterjee, H. Jang, and J. Bokor, "Analysis of ultrafast magnetization switching dynamics in exchange-coupled ferromagnet–ferrimagnet heterostructures," *J. Magn. Magn. Mater.*, vol. 574, 2023.
6. S. Mondal, D. Polley, A. Pattabi, J. Chatterjee, D. Salomoni, L. Aviles-Felix, A. Olivier, M. Rubio-Roy, B. Diény, L. D. B. Prejbeanu, R. Sousa, I. L. Prejbeanu, and J. Bokor, "Single-shot switching in Tb/Co-multilayer based nanoscale magnetic tunnel junctions," *J. Magn. Magn. Mater.*, vol. 581, 2023.
7. Y. C. Lin, Z. Mutlu, G. Borin Barin, Y. Hong, J. P. Llinas, A. Narita, H. Singh, K. Müllen, P. Ruffieux, R. Fasel, and J. Bokor, "Scaling and statistics of bottom-up synthesized armchair graphene nanoribbon transistors," *Carbon*, vol. 205, pp. 519-526, 2023-03-01 2023.
8. J. Engel, K. Jhuria, D. Polley, T. Lühmann, M. Kuhrke, W. Liu, J. Bokor, T. Schenkel, and R. Wunderlich, "Combining femtosecond laser annealing and shallow ion implantation for local color center creation in diamond," *Appl. Phys. Lett.*, vol. 122, 2023.
9. D. Polley, A. Pattabi, J. Chatterjee, S. Mondal, K. Jhuria, H. Singh, J. Gorchon, and J. Bokor, "Progress toward picosecond on-chip magnetic memory," *Applied Physics Letters*, vol. 120, no. 14, p. 140501, 2022.
10. S. Mondal, Y. Lin, D. Polley, C. Su, A. Zettl, S. Salahuddin, and J. Bokor, "Accelerated Ultrafast Magnetization Dynamics at Graphene/CoGd Interfaces," *ACS Nano*, vol. 16, no. 6, pp. 9620-9630, 2022-06-28 2022.
11. G. Borin Barin, Q. Sun, M. Di Giovannantonio, C. Z. Du, X. Y. Wang, J. P. Llinas, Z. Mutlu, Y. Lin, J. Wilhelm, J. Overbeck, C. Daniels, M. Lamparski, H. Sahabudeen, M. L. Perrin, J. I. Urgel, S. Mishra, A. Kinikar, R. Widmer, S. Stolz, M. Bommert, C. Pignedoli, X. Feng, M. Calame, K. Müllen, A. Narita, V. Meunier, J. Bokor, R. Fasel, and P. Ruffieux, "Growth Optimization and Device Integration of Narrow-Bandgap Graphene Nanoribbons," *Small*, p. 2202301, 2022-06-17 2022.
12. M. Guevara De Jesus, Z. Xiao, M. Goiriena-Goikoetxea, R. V. Chopdekar, M. K. Panduranga, P. Shirazi, A. Acosta, J. P. Chang, J. Bokor, G. P. Carman, R. N. Candler, and C. Lynch, "Magnetic state switching in FeGa microstructures," *Smart Materials and Structures*, vol. 31, p. 035005, 2022/01/25 2022.

Curriculum Vitae - Jeffrey Bokor

13. Z. Xiao, C. Lai, R. Zheng, M. Goiriena-Goikoetxea, N. Tamura, C. T. Juarez, C. Perry, H. Singh, J. Bokor, G. P. Carman, and R. N. Candler, "Localized strain profile in surface electrode array for programmable composite multiferroic devices," *Appl. Phys. Lett.*, vol. 118, p. 182901, 2021-05-03 2021.
14. P.-C. Shen, C. Su, Y. Lin, A.-S. Chou, C.-C. Cheng, J.-H. Park, M.-H. Chiu, A.-Y. Lu, H.-L. Tang, M. M. Tavakoli, G. Pitner, X. Ji, Z. Cai, N. Mao, J. Wang, V. Tung, J. Li, J. Bokor, A. Zettl, C.-I. Wu, T. Palacios, L.-J. Li, and J. Kong, "Ultralow contact resistance between semimetal and monolayer semiconductors," *Nature*, vol. 593, pp. 211-217, 2021.
15. Z. Mutlu, J. P. Llinas, P. H. Jacobse, I. Piskun, R. Blackwell, M. F. Crommie, F. R. Fischer, and J. Bokor, "Transfer-Free Synthesis of Atomically Precise Graphene Nanoribbons on Insulating Substrates," *ACS Nano*, vol. 15, p. 2635, Jan 25 2021.
16. Z. Mutlu, P. H. Jacobse, R. D. McCurdy, J. P. Llinas, Y. Lin, G. C. Veber, F. R. Fischer, M. F. Crommie, and J. Bokor, "Bottom-Up Synthesized Nanoporous Graphene Transistors," *Adv Funct Mater*, vol. 31, p. 2103798, 2021.
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EXHIBIT D

Sanjay Banerjee is the Cockrell Family Regents Chair Professor of Electrical and Computer Engineering at the University of Texas at Austin. He received his B.Tech from the Indian Institute of Technology, Kharagpur, and his M.S. and Ph.D. from the University of Illinois at Urbana-Champaign in 1979, 1981 and 1983 respectively, all in electrical engineering. As a Member of the Technical Staff, Corporate Research, Development and Engineering of Texas Instruments Incorporated from 1983-1987, he worked on polysilicon transistors and dynamic random access trench memory cells used by Texas Instruments in the world's first 4Megabit DRAM. He has been Assistant Professor (1987-90), Associate Professor (1990-93), and Professor (1993-) at The University of Texas at Austin. He has over 1000 archival refereed publications/talks, 10 books/chapters, and 35 U.S. patents, and has supervised over 80 Ph.D. and 70 MS students. His students have received 15 Best Papers Awards at various conferences, and he has presented over 150 invited talks. He received the Engineering Foundation Advisory Council Halliburton Award, 1991, the Texas Atomic Energy Fellowship (1990-1997), Cullen Professorship (1997-2001) and the Hocott Research Award from UT Austin (2007). He has won the SIA/SRC University Researcher Award (2017), IEEE Grove Award (2014), Distinguished Alumnus Award, IIT (2005), Industrial R&D 100 Award (2004), ECS Callinan Award (2003), IEEE Millennium Medal (2000), NSF Presidential Young Investigator Award (1988), and several Best Paper Awards and SRC Inventor Recognition Awards. He was a Distinguished Lecturer for IEEE Electron Devices Society, and the General Chair of the IEEE Device Research Conference, 2002. He is a Fellow of IEEE, APS, AAAS and NAI. He is active in the areas of beyond-CMOS nanoelectronic transistors based on 2D materials and spintronics, fabrication and modeling of advanced MOSFETs, and solar cells.

Sanjay Kumar Banerjee

Current Position: Cockrell Family Regents Chair in Electrical and Computer Engineering, 1999-

Education: University of Illinois, PhD (Electrical Engineering), 1983
University of Illinois, MS (Electrical Engineering), 1981
Indian Institute of Technology at Kharagpur, India, B. Tech (Electronics), 1979

Professional Engineer: Texas

Previous Positions: Director, Microelectronics Research Center, 1999-2024
Cullen Trust Endowed Professorship in Engineering, 1997-2001
University of Texas, Assoc. Director, Microelectronics Research Center, 1996-99
University of Texas, Professor, September 1993-
University of Texas, Associate Professor, September 1990- August 1993
University of Texas, Assistant Professor, September 1987- August, 1990
Texas Instruments, Corporate R& D, Member of Technical Staff, 1983-Aug. 1987

Honors and Awards:

Fellow, National Academy of Inventors (2021)
SIA/SRC University Research Award (2017)
IEEE Andrew Grove Award (2014)
Fellow of American Association for Advancement of Science (2007)
Hocott Research Award, Univ. of Texas, 2007
Fellow, American Physical Society, 2006
Distinguished Alumnus Award, IIT, 2005
Industrial R&D 100 Award (with R.Singh) 2004
Electrochemical Society Thomas D. Callinan Award, 2003
IEEE Millennium Medal, 2001
SRC Inventor Recognition Awards, 1994, 2000, 2009
Best Paper Awards (15) @ ISSCC, SRC TECHCON, DARPA, DRC
Who's Who Listings (Marquis)
Cullen Professorship, Univ. of Texas, 1997- 2001
Distinguished Lecturer, IEEE Electron Devices Society (1997-2003), Adcom
Member till 1998
Fellow of IEEE, 1996
Engineering Foundation Advisory Council Halliburton Award, 1991
Texas Atomic Energy Centennial Fellowship, 1990-97
NSF Presidential Young Investigator Award, 1988
Jagadis Bose and National Science Talent Search Scholarships, India, 1974-79
Institute Medal; &Swapan Saha Prize for Highest Ranking UG (ECE), I.I.T., 1979
Phi Kappa Phi

Professional Society and Major Government Committees:

Technical Advisory Board: Applied Novel Devices (current) AstroWatt, DSM Semiconductors, Cambrios, Nanocoolers Inc., BeSang Memories, Organic ID and ITU Ventures; Gerson Lehmann Group, NY; Austin Community College; Asia Pacific IIIT; Rochester Institute of Technology, HSMC Foundry

IEEE Dan Noble Award Committee, 2010-13 (Chair, 2012-13)

Congressional round-table panel member on nanotechnology, Feb. 2008

Member on International Technology Roadmap for Semiconductors

Siemens Westinghouse Science Talent Contest Judge, 2003

Morgan & Claypool Publishers, Lectures in Electronic Materials & Devices, Series Editor

SISPAD, Program Committee, 2005-6

Electrochemical Society Symposium on SiGe, Program Committee, 2004

IRPS, Program Committee, 2005

12th Int. Workshop on Physics of Semiconductor Devices, Int. Advisory Committee

Int. Advisory Committee, Int. Conf. on MEMS and Nanotechnology, IIT, 2005

Program Committee, International SiGe Technology and Device Meeting, 2004-2012

IEEE Device Research Conference Technical Program Chair, 2000-01, General Chair, 2001-02

Editorial Board, Elsevier Science, 2001

IEDM Program Committee, Modeling and Simulation, Session Chair, 2001-03

ECS Session Chair, Toronto, Canada, May 2000

Program Committee, IEEE Int. Conf. Communications, Computers, Devices, Kharagpur, 2000.

IEEE Device Research Conference Program Committee/Local Arrangements Chair, 1999-2000

NSF Workshop Co-Organizer for "Front and Back-end Processes", Austin, TX 1999

Eleventh Int. Ion Implant Tech. Meet. Program Committee and Publications Chair, 1995-1996.

IEEE Symposium on VLSI Technology, Committee Member, 1992-98

NSF Workshop Organizer for "Silicon-Germanium Devices", Austin, TX 1999

IEEE University Government Industry Microelectronics Symp., General Chairman, 1994-1995

IEEE International Electron Devices Meeting, (Device Technology/ Session Chair: 1989-90)

IEEE Conf. on Electromagnetic Field Computation, Chair Comp. in Electron Dev., CA, 1992

Panel Member, SRC Conference on Integration of Novel Processes, 1991

Sponsored Research:

- Grant title:** "Three-Dimensional IC Technology,"
Co-Principal Investigator: S. Banerjee
Other Investigators: D.L. Kwong
Sponsoring Agency: Texas Advanced Technology Program
Duration: June 1988-August 1990.
- Grant title:** "High Speed Devices and VLSI Structures by Laser-Enhanced Epitaxy,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Advanced Technology Program
Duration: June 1988-August 1990.
- Grant title:** "Optoelectronic Devices by Photo-enhanced Chemical Vapor Deposition,"
Principal Investigator: S. Banerjee
Sponsoring Agency: National Science Foundation PYI
Duration: August 1988- July 1993.
- Grant title:** "GaAs-on-Si MESFET Modeling,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Instruments, Inc.
Duration: December 1988- August 1989.
- Grant title:** "Understanding and Modeling of Unit Processes"
Co-Investigator: S. Banerjee
Other Investigators: W. Adcock (PI), A. Tasch (Co-PI), I. Trachtenberg (Co-PI),
D. Kwong, J. Lee, T. Edgar and J. Ekerdt
Sponsoring Agency: SEMATECH and SRC
Duration: December 1988- August 1993.
- Grant title:** "RPCVD Epitaxial Silicon and Insulators for Use in 3-D CMOS Integrated Circuits,"
Co-Investigator: S. Banerjee
Other Investigators: A. Tasch (P.I.), A. Cowley and R. Jones
Sponsoring Agency: Office of Naval Research
Duration: Sept. 1987- March 1990.
- Grant title:** "Ballistic and Quantum Transport in Si Devices at Cryogenic Temperatures"
Principal Investigator: S. Banerjee
Other Investigators: J. Lee
Sponsoring Agency: Texas Advanced Technology Program
Duration: November 1989- November 1991.
- Grant title:** "Polysilicon Transistor Modeling,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Motorola
Duration: September, 1991-August, 1993.

- Grant title:** "Acquisition of High Resolution Transmission Electron Microscope,"
Principal Investigator: L.Rabenberg
Other Investigators: S.Banerjee, J.Goodenough, A.Heller, P.Ho and A.Manthiram
Sponsoring Agency: National Science Foundation
Duration: 10/92-10/93
- Grant title:** "Atomic Layer Epitaxy of Group IV Semiconductors,"
Co-Principal Investigator: S.Banerjee
Other Investigators: A.Tasch (P.I.), A.Cowley, J.Ekerdt and R.Jones
Sponsoring Agency: Office of Naval Research
Duration: February 1991-August 1996.
- Grant title:** "Materials and Bulk Processes"
Co-Investigator: S.Banerjee
Other Investigators: A.Tasch (PI), D.Kwong, J.Lee
Sponsoring Agency: SRC/ SEMATECH
Duration: September 1993- August 1998.
- Grant title:** "Synthesis, Growth and Analysis of Electronic Materials,"
Co-Investigator: S.Banerjee
Other Investigators: J.White (P.I) and 11 others from ECE, Chemistry and Physics
Sponsoring Agency: National Science
Duration: March 1991- March, 1996.
- Grant title:** "Transport in MOSFETs"
Principal Investigator: S.Banerjee
Sponsoring Agency: Motorola
Duration: August, 1993-August, 1994.
- Grant title:** "Flash EEPROMs"
Principal Investigator: S.Banerjee
Sponsoring Agency: AMD
Duration: May, 1993-December, 1996.
- Grant title:** "LDO Thin Film Transistors"
Principal Investigator: S.Banerjee
Sponsoring Agency: Micron
Duration: March, 1993- April, 1995.
- Grant title:** "Ultra Shallow Junction Technology"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: January 1994- December 1996.

- Grant title:** "SIMS Analysis of Polysilicon-on-Silicon"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1994- August 1995.
- Grant title:** "RTP Implant Monitors"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1995- August 1996.
- Grant title:** "Ultra-shallow Junction Formation and 2-D Dopant Profiling"
Principal Investigator: S.Banerjee
Other Investigators: K.Shih
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 1996- December 1997.
- Grant title:** "Analysis of Deep Submicron MOSFETs"
Principal-Investigator: S.Banerjee
Other Investigators: A.Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1998- September 1999.
- Grant title:** "Unrestricted Grant"
Principal Investigator: S.Banerjee
Sponsoring Agency: Various Donors
Duration: No expiration
- Grant title:** "Synthesis, Growth and Analysis of Electronic Materials,"
Co-Investigator: S.Banerjee
Other Investigators: J.White (P.I) and 11 others from ECE, Chemistry and Physics
Sponsoring Agency: National Science Foundation STC
Duration: March 1996- February, 2002.
- Grant title:** "Ultra-shallow Junction Process Integration"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1997- December 2001.
- Grant title:** "Si and Ge Thin Film CVD, Modeling and Control"
Co-Principal Investigator: S.Banerjee
Other Investigators: J.Ekerdt (P.I.), M.Downer, I.Trachtenberg; Univ. of Wisconsin
Sponsoring Agency: Dept. of Defense-MURI
Duration: July 1995-July 2000

- Grant title:** "Ultra-shallow Junction Technology"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 1998- August 2000.
- Grant title:** "Channel Engineering in Si-Ge-C MOSFETs "
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1997- September 2000.
- Grant title:** "Advanced Annealing"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 2000- December 2001.
- Grant title:** "Quantum Transport in Heterostructure MOSFETs "
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1999- September 2002.
- Grant title:** "Front End Processing"
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch, D. Kwong, J. Lee
Sponsoring Agency: SRC/ SEMATECH
Duration: April 1998- March 2001.
- Grant title:** "Vertical Si-Ge-C MOSFETs "
Principal-Investigator: S. Banerjee
Sponsoring Agency: Semiconductor Research Corporation
Duration: September 2000- August 2003.
- Grant title:** "Compact Modeling of Gate Current"
Principal-Investigator: S. Banerjee
Other Investigators: F. Register
Sponsoring Agency: Semiconductor Research Corporation
Duration: July 2000- September 2003.
- Grant title:** "Ion Implantation Modeling"
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: July 2000- September 2001.

Grant title: "Front End Processing"

Principal-Investigator: S.Banerjee

Other Investigators: D.Kwong, J.Lee, F.Register

Sponsoring Agency: SRC/ SEMATECH

Duration: April 2001- March 2003.

Grant title: "MARCO Focus Center on Device Structures"

PI at UT: S.Banerjee

Other Investigators: D.Kwong (with MIT, Stanford, UC Berkeley)

Sponsoring Agency: DARPA/SRC

Duration: Award announced February 2001 (3 year contract)

Grant title: "SiGe Flash EEPROMS with Quantum Dot Gates"

Principal Investigator: S.Banerjee

Sponsoring Agency: Texas Higher Education Coordinating Board

Duration: January 2002- December 2003.

Grant title: "MARCO Focus Center on Device Structures"

Principal-Investigator: S.Banerjee

Other Investigators: D.Kwong (with MIT, Stanford, UC Berkeley)

Sponsoring Agency: DARPA/SRC

Duration: Sept. 2003 (3 year contract)

Grant title: "High mobility Ge-channel MOSFETs "

Principal-Investigator: S.Banerjee

Sponsoring Agency: Semiconductor Research Corporation

Duration: September 2003-August 2006.

Grant title: "Monte Carlo and Quantum transport "

Principal-Investigator: S.Banerjee

Co-PI: L.F.Register

Sponsoring Agency: Semiconductor Research Corporation

Duration: September 2003-August 2006.

Grant title: "NIRT on Quantum Dot Memories "

Principal-Investigator: S.Banerjee

Other Investigators: J.Ekerdt, F.Register, G.Hwang

Sponsoring Agency: NSF

Duration: September 2003-August 2007.

Grant title: "High mobility Ge-channel MOSFETs "

Principal-Investigator: S.Banerjee

Sponsoring Agency: Texas Higher Education Coordinating Board

Duration: January 2004-Dec. 2005.

- Grant title:** "Advanced Materials Research Center"
Principal-Investigator: S.Banerjee
Other Investigators: 15 others
Sponsoring Agency: Texas
Duration: January 2004- Dec.2005
- Grant title:** "Advanced Processing and Prototyping Center"
Principal-Investigator: S.Banerjee
Other Investigators: 18 others
Sponsoring Agency: DARPA
Duration: 2005- Dec.2006
- Grant title:** "SiGe Nanostructures"
Co-Principal-Investigator: S.Banerjee,
Other Investigators: R.Huang
Sponsoring Agency: DOE
Duration: 2006- Dec.2009
- Grant title:** "Dopant Diffusion Modeling"
Principal-Investigator: S.Banerjee,
Other Investigators: G.Hwang
Sponsoring Agency: SRC
Duration: 2006- Dec.2009
- Grant title:** "NNIN "
Site Director: S.Banerjee
Sponsoring Agency: NSF
Duration: January 2004- August.2015
- Grant title:** "MARCO MSD Focus Center "
UT PI: S.Banerjee
Sponsoring Agency: DARPA/SRC
Duration: Sept. 2007-2012
- Grant title:** "CERA"
Principal-Investigator: S.Banerjee
Co-PIs: F.Register, R.Ruoff, E.Tutuc, A.Macdonald, D.Akinwande
Sponsoring Agency: DARPA/IBM
Duration: Sept. 2007-2012
- Grant title:** "NASCENT ERC "
Principal-Investigator: Bonnecaze, Sreenivasan
Banerjee (Device Thrust co-Leader)
Sponsoring Agency: NSF
Duration: January 2013- Feb.2018

- Grant title:** "Bay Area PV Consortium led by Stanford/Berkeley"
Sponsoring Agency: DOE
Duration: Sept. 2015- August.2017
- Grant title:** "South West Academy of Nanoelectronics"
Director: S.Banerjee,
Other Investigators: F.Register, A.MacDonald and 15 others from 6 schools
Sponsoring Agency: SRC-NRI
Duration: 2006- Dec.2018
- Grant title:** "NSF-NNCI "
Site Director: S.Banerjee
Sponsoring Agency: NSF
Duration: Sept. 2015- August.2025
- Grant title:** "MURI-Room Temperature Polariton Condensates "
Principal-Investigator: Hui Deng (Michigan),
Co-PIs: S.Banerjee and 4 others
Sponsoring Agency: DoD MURI
Duration: Sept. 2017- August.2023
- Grant title:** "Soft-FET "
PI: J.Kulkarni, S.Banerjee (co-PI)
Sponsoring Agency: NSF
Duration: Sept. 2018- August.2022
- Grant title:** "EFRI- sub-contract from UT Dallas"
PI: S.Banerjee
Sponsoring Agency: NSF
Duration: Sept. 2018- August.2022
- Grant title:** "3D CMOS using CVD TMD"
PI: S.Banerjee
Sponsoring Agency: SRC
Duration: Jan. 2021- Dec.2023
- Grant title:** "BEOL Transistor"
PI: S.Banerjee
Sponsoring Agency: TI, \$85k
Duration: June. 2024- Dec.2025
- Grant title:** "TMD FETs"
PI: S.Banerjee
Sponsoring Agency: Samsung, \$720k

Duration: Sept. 2023- Aug.2026

Grant title: "ALE Modeling"

PI: S.Banerjee; co-PI Gyeong Hwang (ChemE)

Sponsoring Agency: Samsung, \$660k

Duration: Sept. 2023- Aug.2026

Grant Title: " Applied Novel Devices- Power FETs"

PI: S.Banerjee. co-PIs, Alex Huang, Alex Hanson

Sponsoring Agency: Army, \$368,753

Duration: Aug. 2023- Aug.2025

Ph.D. supervision:

Keun Park, 1991
Ting Hsu, 1991
Sean Lian, 1991
Shubneesh Batra, 1992
Sittampalam Yoganathan, 1992
Burt Fowler, 1992
David Kinosky, 1993
Surya Bhattacharaya, 1993
Brain Li, 1994
Le-tien Jung, 1994
Avinash Mahajan, 1994
Chung-you Hu, 1995
Indrajit Manna, 1995
Akif Sultan, 1996
Dean Samara, 1997
Soji John, 1998
Jacob Liu, 1999
Rajan Sharma, 1999
Ed Quinones, 1999
David Kencke, 2000
Christine Ouyang (with Tasch), 2000
Xiangdong Chen, 2001
Taehoon Kim, 2001
Siva Mudanai (with Tasch), 2001
Geng Wang (with Tasch), 2001
Yang Chen (with Tasch), 2001
Xin Wang, 2002
Hong-Jyh Li, 2002
Sung –Bo Hwang (with Edgar), 2002
Di Li (with Tasch), 2002
Yang-Yu Fan (with Register) 2002
Zhonghai Shi, 2002
Tat Ngai, 2002
Dong-Won Kim, 2003
Xiao Chen (with Rabenberg), 2003
Puneet Kohli, 2003
David Onsongo, 2003
Kartik Jayanarayan, 2004
Tongsheng Xia, (with Register), 2005
Taras Kirichenko (with Hwang), 2005
James Chen, 2005
Swaroop Ganguly (with MacDonald), 2006
Fei Lei (with Register), 2006
Li Lin (2006)
Sagnik Dey (2006)
Xiangdong Fan (with Register) 2006
Yueran Liu, 2006
David Kelly, 2006
Xiaofeng Fan (with Register) 2006
Bahman Ghosh (with Register) 2007

Sachin Joshi, 2007
Joy Sarkar, 2007
Joseph Donnelly, 2009
Davood Shahrjerdi, 2008
Shan Tang, 2008
Rownak Zaman, 2008
Ning Kong, 2009
Yonghyun Kim, 2010
Hai Liu 2010
Dipanjan Basu (with Register), 2010
Tackhwi Lee, 2010
Se Hoon Lee, 2011
Ferdousi Fahmida, 2011
Jamil Mustafa, 2011
John David (with Register) 2011
Jung Hwan Yum, 2012
Seyoung Kim, (with Tutuc) 2012
Chang, Jiwon, (with Register) 2013
Priyamvada Jadaun, (with Register) 2013
Yujia Zhai, (with Willson) 2013
Michael Ramon (with Akinwande) 2013
Emmanuel Oneyagam, 2014
Sayan Saha, 2015
Urmimala Roy (with Register) 2015
Jason Mantey, 2015
Donghyi Koh (2016)
Chris Corbet (with Tutuc) 2016
Sangwoo Kang (2016)
William Hsu (2016)
Andreas Hsieh (2017)
Tanuj Trivedi (with Neikirk) (2017)
Atresh Sanne (2017)
Tanmoy Pramanik (with Register) (2018)
Jaehyun Ahn (2018)
Hema Movva (with Tutuc) (2018)
Harry Chou (2018)
Amritesh Rai (2019)
Rik Dey (with Register) (2019)
Omar Mohammed (2019)
Aqyan Bhatti (with Register) (2020)
Sayema Chowdhury (2022)
Teja Subrahmanyam (with Kulkarni) 2023
Moonkyu Song (2024)
Matthew Disiena (2025)

M.S.:

D.Bullock, 1990
K.Picone, 1991
J.Shen, 1991
R.Kovelamudi, 1992
S.Krishnan, 1992
M.Lobo, 1992
S.Ngaoram, 1993
A.Khan, 1993
D.Khanderkar, 1993
H.Taufique, 1994
S.Madireddi, 1994
J.Fretwell, 1995
M.Craig, 1995
J.Williamson, 1995
K.Reddy, 1995
J.Damiano, 1995
R.Gupta, 1995
K.Hassan, 1996
A.Lentvorski, 1997
S.Oswal, 1998
C.Seal, 1998
V.Agarwal, 1999
S.Nandan, 1999
S.Ravi, 1999
T.Ngai, 1999
V.Medina, 1999
H.Rahman, 1999
C.Twu, 2000
S. Oak, 2001
G. Shrivastava, 2001
R. Deppensmith, 2002
M. Swaminathan, 2002
L. Lin, 2003
L.Weltzer, 2004
D.Ahmad, 2005
I.Wiedmann, 2005
S. Ramachandran, 2005
A.Nanda 2007
N. Jain, 2008
N.Vora, 2008
K. Varahramyan, 2008
S. Kaur, 2010
Stephen Szczepaniak, 2015
Jessica Depoy, 2020
Alexander Klatt, 2021
Nick Pronin, 2021
Isaac Bodeman, 2022

Postdocs: Samit Ray (Chair Physics, IIT, Dean S N Bose Institute), Mark Loewe (IBM), Amitava Das (startup), Sabrina Grannan (NASA JPL), Freek Prins (Germany), Chuanbin Mao (Chair Prof. Oklahoma), Bhagawan Sahu (Global Foundries), Mathew Gilbert (Assoc. Prof. UIUC), Donwan.Ahn (Korea), Domingo Ferrer (IBM), Aparna Gupta (IIT), Samaresh Gucchait (Asst. Prof., Howard), Sushant Sonde (Argonne), SeHoon Shin (Samsung), Priyamvada Jadaun (Cornell/IMEC), Marylene Palard, Bahniman Ghosh (IIT), Rudresh Ghosh (NovaCentrix), Victor Chi, Sungkyu Kwon, Seung Heon Shin (Samsung), Anupam Roy (Asst. Prof. IIT), Sarmita Majumder (Sheetak), Ansh Gupta (IMEC), Nupur Navlakja, Prakriti Neha, Nilesh Pandey, Rajveer Jha

Current Post-docs: Nilesh Pandey, Rajveer Jha

In progress (Ph.D):

Chris Luth, Ryan Schalip, Siyu Wu, Sunny Bhakta, Hadi Darkhaneh, Jatin Singh, Hongming Zhang, Ashkan Aminian, Luke Sloan, Pranav Rama, John Komoll, Nicholas Watanabe, Blake Broussard, Justin Tullos

Books and Invited Book Chapters:

1. Solid State Electronic Devices, 5th Ed. (2000), 6th Ed. (2005), 7th Ed. (2015), Prentice-Hall by B.Streetman and S.Banerjee
2. Effect of Surface Nitridation on the Electrical Characteristics of Germanium High- κ /Metal Gate Metal-Oxide-Semiconductor Devices, D. Q. Kelly, J. J.-H. Chen, S. Guha, and S. K. Banerjee. Invited Book chapter, Springer, 2007.
3. SiGe HFETs, S.Banerjee, The Silicon Heterostructure Handbook, 2005, Edited by John Cressler.
4. High-k Gate Dielectrics, Y.Fan. S.Mudanai, L. Register and S.Banerjee, 2003
5. Device Miniaturization and Simulation, S.Banerjee and B.Streetman in ULSI Devices, John Wiley, 2000 (C.Chang and S.Sze editors)
6. Dopant Diffusion, S.Banerjee in Handbook of Semiconductor Manufacturing Technology, Marcel Dekker, 2000, 2006 (Y.Nishi, B.Doering and J.Kilby editors).
7. Silicon-germanium Devices, S.Banerjee, Elsevier, 2001.
8. Novel 3D CMOS, S.Dey and S.Banerjee, Solid State Electronics Trends, 2009
9. X. Mou, L. F. Register and S. K. Banerjee, "Ultra-low-power pseudospintronics devices via exciton condensation in coupled two-dimensional material systems," in Nanoscale Materials and Devices for Electronics, Photonics and Solar Energy, Ed. Stephen Goodnick, Anatoli Korkin and Robert Nemanich, Springer, 2015
10. D. Reddy, L. F. Register and S. K. Banerjee, "Bilayer pseudoSpin Field Effect Transistor (BiSFET)" in "Beyond CMOS Logic Switches," T.-J. King and K. Kuhn, Eds., Cambridge: Cambridge Univ. Press, 2015.
11. Devices and defects in two-dimensional materials: outlook and perspectives, A Rai, A Roy, A Valsaraj, S Chowdhury, D Taneja, Y Wang, LF Register, SK Banerjee Defects in Two-Dimensional Materials, 339-401 (Invited Book Chapter) 2021.

1. US20200365464 11/19/2020
Catalyst influenced chemical etching for fabricating three-dimensional sram architectures and optical waveguides, S.Sreenivasan, A.Mallavarapu, J.Kulkarni, M.Watts and S.Banerjee
2. US10,121,962 11/06/2018
Method for fabricating magnetic solid state devices. L.F.Register, B.Ghosh, R.Dey and S.Banerjee
3. US9,825,218 11/21/2017
Transistor that employs collective magnetic effects thereby providing improved energy efficiency, A.MacDonald, L.F. Register, E. Tutuc, I. Sodemann, H. Chen, X. Mou, S. Banerjee
4. US8,709,892 4/29/2014
Nanoparticles in a flash memory using chaperonin proteins, C.Mao, S.Tang and S.Banerjee
5. US8,629,427 1/14/2014
Topological insulator-based field-effect transistor, S.Banerjee, L.Register, A.MacDonald, B.Sahu, P.Jadaun and J.Chang
6. US8,263,967 9/11/2012
Bi-layer pseudo-spin field-effect transistor, S.Banerjee, L.Register, A.MacDonald, D.Reddy, E.Tutuc
7. US8,198,707 6/12/2012
Establishing a uniformly thin dielectric layer on graphene in a semiconductor device without affecting the properties of graphene, L.Colombo, S.Banerjee, S.Kim, E.Tutuc
8. US8188460 5/29/2012
Bi-layer pseudo-spin field-effect transistor, S.Banerjee, L.Register, A.MacDonald, D.Reddy, E.Tutuc
9. US8,008,649 8/30/2011
Incorporating gate control over a resonant tunneling structure in CMOS to reduce off-state current leakage, supply voltage and power consumption, L.Register and S.Banerjee
10. US6,744,083 6/1/2004
Submicron MOSFET having asymmetric channel profile, X.Chen and S.Banerjee
11. US6,420,219 7/16/2002
Thin film transistor and method, S.Batra, M.Manning, S.Banerjee and J.Damiano
12. US06320202 11/20/2001
Bottom gated thin film transistors comprising Ge in a channel region, S.Banerjee and S.Batra
13. US06,319,799 11/20/2001
High mobility heterojunction transistor and method, Q.Ouyang, A.Tasch and S.Banerjee
14. US06,313,486, 11/06/2001
Floating gate transistor having silicon germanium channel layer, D.Kencke and S.Banerjee
15. US06,313,487 11/06/2001
Vertical channel floating gate transistor having SiGe channel layer, D.Kencke and S.Banerjee
16. US06214652 4/10/2001
Thin film transistor and method of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
17. US06,200,839 3/13/2001

- Methods of making thin film transistors, S.Batra, M.Manning, S.Banerjee and L.Jung
18. US06166398 12/26/2000
Thin film transistors, S.Batra, M.Manning, S.Banerjee and L.Jung
 19. US06017782 01/25/2000
Thin film transistor and method of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
 20. US05985703 11/16/1999
Method of making thin film transistors, S.Banerjee
 21. US05977560 11/02/1999
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June 2025

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Byoung Hak Hong, Wookhyun Kwon, Branch T. Archer, Leonard F. Register, IEEE,
Sanjay K. Banerjee, and Jiwon Chang, IEEE EDL Dec. 2025.

EXHIBIT E

Suman Datta

Joseph M Pettit Endowed Chair Professor
Georgia Research Alliance (GRA) Eminent Scholar
School of Electrical and Computer Engineering
Georgia Institute of Technology

2360 Klaus Advanced Computing Building; 266 Ferst Drive, Atlanta, GA 30332

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Biography

Suman Datta received the B.Tech degree in Electrical Engineering from the Indian Institute of Technology, Kanpur, India and the Ph.D degree in Electrical and Computer Engineering from University of Cincinnati, Cincinnati, Ohio. He joined Georgia Tech's School of Electrical and Computer Engineering (ECE) as the Joseph M. Pettit Chair and Georgia Research Alliance (GRA) Eminent Scholar in the Fall of 2022. From 2016 till 2022, he was the Stinson Endowed Chair Professor of Nanotechnology in the Department of Electrical Engineering at the University of Notre Dame, Notre Dame, Indiana, where he directed research in the Nanoelectronic Devices and Circuits Laboratory. He was the Director of the multi-university \$40.4M SRC and DARPA sponsored Applications and Systems-driven Center for Energy-efficient integrated NanoTechnologies (ASCENT) till Dec 2022. In addition, he was the Director of the SRC and NSF sponsored Center for Extremely Energy Efficient Collective Electronics (EXCEL). His research involves heterogeneous computing, brain-inspired computing, in-memory/near-memory computing and cryogenic computing using advanced CMOS and beyond CMOS devices. Prior to Notre Dame, he was a Professor of Electrical Engineering at The Pennsylvania State University, University Park, from 2007 to 2015. From 1999 till 2007, he was in the Advanced Transistor Group at Intel Corporation, Hillsboro, where he pioneered several generations of high-performance logic transistor technologies including high-k/metal gate, Tri-gate (FinFET) and non-silicon channel CMOS transistors that have been deployed in Intel microprocessors for high-performance cloud and client computing applications.

Prof. Datta has co-authored 12 book chapters and over 510 refereed journal and conference papers, holds 185 United States patents and has presented numerous invited and keynote talks. His work has won 3 best paper awards (DRC 2010 and 2011), Test-of-time award (VLSI Symposium 2020), best paper nominations (IEDM 2016, 2018) and received over 43,475 citations (*h-index* = 106). He is a recipient of the Intel Achievement Award (2003), the Intel Logic Technology Quality Award (2002), the Penn State Engineering Alumni Association (PSEAS) Outstanding Research Award (2012), the SEMI Award for North America (2012), the PSEAS Premier Research Award (2015), and the SIA-SRC University Researcher Award (2024). Datta recently received the 2026 IEEE Andrew S. Grove Award *for contributions to and leadership in high-performance and energy-efficient logic transistor research* – the IEEE Technical Field Award presented by the IEEE Board of Directors for outstanding contributions to solid-state devices and technology. Datta served as the General Chair of premier IEEE conferences (DRC and IEDM) and served on the Technical Program Committee of top-tier IEEE conferences (VLSI Technology Symposium, IRPS, VLSI-TSA etc.). In 2013, he was named Fellow of the Institute of Electrical and Electronics Engineers (IEEE) *for his contributions to high-performance advanced silicon and compound semiconductor transistor technologies*. In 2016, he was named Fellow of the National Academy of Inventors (NAI) *in recognition of his inventions that have made a tangible impact on quality of life, economic development and the welfare of society*.

Education

PhD, Electrical and Computer Engineering Dept., University of Cincinnati, OH, September 1999

Bachelors, Electrical Engineering Dept., Indian Institute of Technology, Kanpur, India, June 1995

Experience

Last updated, Mar 2026

Joseph M Pettit Endowed Chair in Advanced Computing and Professor (2022 – Now)

School of Electrical and Computer Engineering, Georgia Institute of Technology, GA

Stinson Endowed Chair Professor (2015 – 2022)

Department of Electrical Engineering, University of Notre Dame, IN

Director, Applications and Systems Driven Center for Energy-Efficient NanoTechnology (2018-2023)

DARPA/ SRC supported multi-university \$40.4M center with 29 PIs, 14 schools, 120 researchers

Director, Center for Extremely Energy-Efficient Collective Electronics (2016 - 2021)

NSF/ SRC supported multi-university \$4.49M center with 10 PIs, 5 schools, 15 researchers

Professor (2011 – 2015)

Department of Electrical Engineering, Penn State University, PA

Monkowski Associate Professor (2007-2011)

Department of Electrical Engineering, Penn State University, PA

Principal Engineer (2005-2007)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Senior Staff Engineer (2003-2005)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Staff Engineer (2002-2003)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Senior Process Engineer (1999-2002)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Technology CAD (TCAD) Engineer (1999)

Avanti Corporation (now Synopsis), Fremont, CA

Technology Consultant (2008 – 2022)

TSMC (Taiwan Semiconductor Manufacturing Corp)

UMC (United Microelectronics Corp)

Samsung Electronics

Micron

Subject Matter Expert (2020-2023)

Expert Witness for Samsung in the KIPB LLC versus Samsung Electronics Co. (2020)

Expert Witness for Samsung in the Daedalus LLC versus Samsung Electronics Co ITC Case
(2023)

Research Career

Funded Projects

NRI

Last updated, Mar 2026

- Energy Efficient Transistor and Architectures, (06/08-05/11)- Phase 1, **Nanoelectronics Research Institute/Semiconductor Research Corporation (NRI/SRC)**, \$525,920 (Total: \$873,920 including PSU match of \$348,000), (PI with T. Mayer, V. Narayanan, D. Schlom)(40% share)
- Energy Efficient Transistor and Architectures, (06/11 – 12/12) - Phase 1.5, **Nanoelectronics Research Institute/Semiconductor Research Corporation (NRI/SRC) and National Institute of Standards and Technology (NIST)**, \$280,000 (PI with T. Mayer)(50% share)
- E2CDA: Extremely Energy Efficient Collective Electronics, (10/16 – 8/31), **National Science Foundation (NSF) and Semiconductor Research Corporation (SRC)**, \$4,290,000 (Center Director and PI) (Candidate's share \$430,000)

NSF

- Ultra-sensitive Magnetic Sensors Integrating the Giant Magnetoelectric Effect with MEMs and Advanced Microelectronics, (10/08-9/11), **National Science Foundation/Division of Electrical, Communications and Cyber Systems (NSF/ECCS)**, \$352,273 (Co-PI with Q. Zhang (PI), EE, PSU and Q. Yang, Radiology, PSU Hershey Medical Center)(50% share)
- EMT/NANO:Co-Exploration of Device and System Architecture for Quantum NanoElectronics,” (09/08-08/11), **National Science Foundation/Division of Computing and Communication Foundation (NSF/CCF)**, \$200,020 (Co-PI with V. Narayanan (PI), EE, PSU)(50% share)
- Collaborative: Mixed Anion and Cation Based Transistor Architecture for Ultra-Low Power Complementary Logic Applications, (10/10-09/13), **National Science Foundation/ Division of Electrical, Communications and Cyber Systems (NSF/ECCS)**, \$472,753 (PI with M. Hudait, EE, Virginia Tech)(50% share)
- MRSEC: Center for Nanoscale Science Supplement titled “Very Low Energy Dissipation Computing using Inter-band Tunneling Injected Non-equilibrium Ballistic Carriers,” (10/10-09/13), **National Science Foundation/Nanotechnology Research Initiative (NSF/NRI)**, \$300,000, (Co-PI with T. Mayer(PI), EE, PSU)(50% share)
- ERC: NSF Nanosystems Engineering Research Center (NERC) for “Advanced Self-Powered Systems of Integrated Sensors and Technologies (ASSIST),” (10/12 – 09/19), **National Science Foundation**, \$18,500,000 (Co-PI and Low Power Nanoelectronics Theme Leader)(Candidate's share \$600,000 to date)
- EXPEDITION: NSF Expedition in Computing for “Visual Cortex on Silicon,” (10/13 – 09/18), **National Science Foundation**, \$10,000,000 (Low Power Nanoelectronics Theme Leader)(Candidate's share \$500,000)

- EFRI 2DARE: Ultra-Low Power, Collective-State Device Technology Based on Electron Correlation in Two-Dimensional Atomic Layers, (9/13 – 12/17), **National Science Foundation**, \$2,000,000 (Co-PI)(Candidate's share \$500,000)
- MRSEC: Center for Nanoscale Science,” (10/13-09/16), **National Science Foundation (NSF)**, \$18,300,000, (Senior Personnel) (Candidate's share \$250,000)
- SemiSynBio: Cardiac Muscle-Cell-Based Coupled Oscillator Networks for Collective Computing (07/18 - 06/21), **National Science Foundation (NSF)**, \$1,125,000 (Co-PI) (Candidate's share \$300,000)
- 2D Defect Zoology, Functionalization, and Application, **National Science Foundation (NSF)** (04/18 - 9/19), \$75,000 (Candidate's share 100%)
- SemiSynBio: Cardiac Muscle-Cell-Based Coupled Oscillator Networks for Collective Computing (07/18 - 06/21), **National Science Foundation (NSF)**, \$1,125,000 (Co-PI) (Candidate's share \$300,000)
- Collaborative Research: DMREF: Accelerated Design, Discovery, and Deployment of Electronic Phase Transitions (ADEPT) (1/24 - 12/27), **National Science Foundation (NSF)**, \$2,000,000 (Co-PI) (Candidate's share \$400,000)
- FuSe: A Reconfigurable Ferroelectronics Platform for Collective Computing (FALCON) (10/23 - 9/26), **National Science Foundation (NSF)**, \$1,290,000 (Co-PI) (Candidate's share \$300,000)
- “EFRI BEGIN OI: Implementation of Dense Associative Memory through Cardiac muscle cell-based Reprogrammable Bio-Oscillatory Neural Networks”, **National Science Foundation**, 09/2024-08/2028, \$400,000

DARPA, NSA, ONR, NIST, DTRA, SRC

- Mixed Anion Arsenide-Antimonide Channel Transistors with High-k Gate Stack, (11/09-10/12) **Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA)**, \$226,000 (PI)
- Architecture-Device Co-Design for Ultra-Low Power High Performance Design, (10/09-09/11), **National Security Agency (NSA)**, \$600,000 (Co-PI with V. Narayanan, EE, PSU)(50% share)
- Correlated Electron Switching Based Tunnel Transistors, (7/11-6/15), **Office of Naval Research (ONR)**, \$1,923,700 (PI with V. Gopalan, R. Engel-Herbert, MSE, PSU, D. Schlom, MSE, Cornell, K. Rabe, Physics, Rutgers)(25% share)
- Development and Demonstration of Next Generation Electronic Warfare Components based on Graphene Technologies, (01/02/12 – 12/31/14), **Office of Naval Research (ONR)**, \$1,280,030 (Co-PI with J. Robinson, Electro-optic Center, PSU)(33% share)

- Ultrafast Spectroscopy in Heterojunction Tunnel Transistors, (10/11 – 9/13), **National Institute of Standards and Technology (NIST)**, \$120,000 (PI)
- Basic Single-Event and Total-Ionizing Dose Mechanisms in Antimony (Sb)-based CMOS Transistors with High-K Dielectric, (4/01/14-3/31/17), **Defense Threat Reduction Agency (DTRA)**, \$1,745,560 (PI with D. McMorro, NRL, K. Saraswat, EE, Stanford U.)(33% share)
- Architecture-Device Co-Design for Ultra-Low Power High Performance Design- Phase 2, (10/12-09/14), **National Security Agency (NSA)**, \$600,000 (Co-PI with V. Narayanan, EE, PSU)(50% share)
- Center for Low Energy Systems (LEAST) FCRP with Notre Dame Univ. (01/13-12/17)
Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA) (total center funding is \$ 30,000,000), Candidate is PI and Theme Leader for the “Quantum Engineered Steep Slope Transistors” \$ 4,000,000 (over 5 years) (25% share)
- Oxide-based Reconfigurable Single-Electron Logic for Beyond CMOS, (10/13 - 9/14) **(Semiconductor Research Corporation/Sematech)** \$62,125 (PI)
- Basic Single-Event and Total-Ionizing Dose Mechanisms in GE/InGaAs-based CMOS Transistors with ALD High-k Dielectric, (09/14 – 08/17) **Defense Threat Reduction Agency (DTRA)**, \$1,045,560 (PI with C. Cress, NRL, K. Saraswat, EE, Stanford U.)(33% share)
- Landau FET Using Mott Hubbard Phase Transition, (1/14 - 12/17) **(Semiconductor Research Corporation)** \$325,000 (PI)
- Ferroelectric Field Effect Transistor with Steep Switching Slope and Non-Volatile Functionality, (1/16 – 12/18) **(Semiconductor Research Corporation)** \$300,000 (PI)
- Orbital Ordering Driven Threshold Switches for Select Devices in 3D X-Point Memories (11/15 – 10/17) **Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA)** \$ 500,000 (PI with S. Gupta, Penn St, and S. Guha, University of Chicago)(50% share)
- ASCENT - Applications and Systems driven Center for Energy-Efficient Integrated NanoTechnologies (01/18 - 12/22) **DARPA / Semiconductor Research Corporation** \$ 38,386,478 (Director and Center Leader) (Candidate’s share: \$1,200,000)
- IMPACT - Innovative Materials and Processes for Accelerated Compute Technologies (IMPACT) Research Center (01/20 – 12/22) **DARPA / Semiconductor Research Corporation** \$ 6,000,000 (PI) (Candidate’s share: \$1,000,000)
- Multi-Bit-per-Cell (MBC) Ferroelectric FET Memory Using Ferroelectric (FE) Superlattice with Anti-Ferroelectric (AFE) Interfacial Coupling, (1/20 – 12/22) **(Semiconductor Research Corporation)** \$255,000 (PI)

- Multi-Component Semiconducting Oxide FETs: Materials-Device Co-Design, Synthesis, NanoFabrication, Characterization and Benchmarking, (1/20 – 12/22) (**Semiconductor Research Corporation**) \$255,000 (PI)
- T-MUSIC: Negative Capacitance Enabled Scaling to Achieve 1 THz Cut-off Frequency Transistors on a CMOS Platform, (0/19 – 3/21) (**DARPA**) \$444,302 (PI)
- IIRM-URA: Interaction with Ionizing Radiation with Matter, University Research Alliance (IIRM-URA) (07/20 – 12/24) (**DTRA**) \$1,872,523 (PI)
- FALCON: Ferroelectric Analog Compute in Memory (11/23 – 4/28) (**DARPA**) \$9,000,000 (PI)
- 3DFEM: Center for 3D Ferroelectric Microelectronics (08/22 – 7/24) (**DOE**) \$240,000 (PI)
- Center for Heterogenous Integration of Micro Electronic Systems (CHIMES) (01/23 – 12/27) (**SRC/DARPA**) \$1,094,092 (PI)
- Center for Processing with Intelligent Storage and Memory (PRISM) (01/23 – 12/27) (**SRC/DARPA**) \$1,089,581 (PI)
- Zero Schottky Barrier Ohmic Contact to Silicon Using Metal Interlayer (01/23 – 12/25) (**SRC**) \$405,000 (PI)
- Advanced Technology Boosters to Enable Low Temperature Logic Technology (LTLT) on a CMOS Platform (01/22 – 10/24) (**DARPA**) \$2,741,301 (PI)
- Atoms-to-Systems Co-Design: Transforming Data Flow to Accelerate Scientific Discovery (09/21 – 10/24) (**DOE**) \$525,000 (Co-PI)
- Machine Learning Enabled High Precision and Fast Thermal Model of Nanoscale Transistor Arrays (9/23-8/24) (**DARPA**) \$500,000 (Co-PI)
- FALCON: Ferroelectric Analog Compute-in-Memory, **DARPA**, 12/2023-8/2026, \$3,000,000 (PI ~ 20% share)

Industry

- Compound Semiconductor Based Heterojunction Tunnel Transistors for Ultra Low Power Logic Applications-Phase 2, (09/01/09-08/31/12), **Intel Corporation**, \$255,000 (PI)
- Ultra-Low Resistance Ohmic Contacts for III-V Digital Logic, (04/01/09-05/01/11), **Intel Corporation**, \$250,000 (Co-PI with S. Mohny, Mat. Sc., PSU)(50% share)
- Post CMOS circuits and architecture, (10/01/10-09/30/13), **Academic Research Office (ARO), Intel Corporation**, \$170,000 (PI)
- Sub-0.4V Logic Circuits with Steep Sub-threshold Slope Inter-band Tunnel FETs-Phase 2, (06/09-06/11) **Intel Corporation**, \$70,000 (PI)

- Supply voltage scalability of III-V based heterojunction tunnel transistors -Phase 1, (09/08-08/09), **Intel Corporation**, \$85,000 (PI)
- Multi-Gate III-V QWFET , (3/1/11-2/28/14), **Global Foundries**, \$165,000 (PI)
- Germanium and III-V Devices, (08/11 – 07/12), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Ultra Low Resistivity Metal Insulator Semiconductor (MIS) Contacts, (10/12 – 09/13), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Reliability Assessment of Highly Scaled High-k Gate Stacks, (10/12 – 09/13), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Variation Study of 3D Transistors, (10/11 – 09/14), **Lam Research**, \$ 75,000 (PI)
- III-V-based Nanowire MOSFET and NanoPillar Tunnel FET for Ultra Low-Power Nanoelectronics, (2/12 – 07/15), **Samsung GRO**, \$ 240,000 (PI)
- Ultra-scaled III-V FinFETs for Next Generation Nanoelectronics (11/15 – 1/17), **Samsung Electronics**, \$ 120,000 (PI)
- 5nm Node Logic Transistor Option for Mobile System on a Chip (“SOC”) (2/16 – 1/17), **Qualcomm**, \$ 50,000 (PI)
- Low Voltage, High Non-Linearity (High ION & Low IOFF) Select Devices for Cross-point Memory Applications (06/17 - 05/20), **Intel Corporation**, \$300,000 (PI)
- Functional Oxide based Cross-point Resistive Devices for Deep Neural Networks (04/17 - 04/19) **IBM Corporation**, \$60,000 (PI)
- Investigation of Oxygen Insertion Channel Field Effect Transistor Architecture for Enhanced Transport and Improved Positive Bias Temperature Instability (4/17 – 2/19) **ATOMERA**, \$117,393 (PI)
- Transport in Gate-All-Around (GAA) Nanowire (NW) Transistors: Impact of NW Shape and Diameter (11/16 - 10/18), **Silicon Valley Community Foundation** \$25,000 (PI)
- Ferroelectric Transistor Technologies for Neuromorphic Computing (03/23 – 3/25) **Versum Materials**, \$600,000 (PI)
- A scalable physical compute engine for energy-based learning and inference (09/21 – 9/24) **Intel Corporation**, \$255,000 (PI)
- Technology Boosters for DRAM Logic Transistors, (5/2024 - 04/2026), **Micron Technology**, \$227,272 (PI)
- Performance and reliability boosters for amorphous oxide semiconductor (AOS) TFT", (5/2024 - 04/2027), **Samsung Electronics**, \$450,000 (PI ~ 50%)

Total number of publications to date:

– **490 total (240 journal, 250 refereed conferences, 186 patents)**

Last updated, Mar 2026

– h-index = **106** with **>43,475** citations (Google Scholar)

Total number of graduate students to date:

Postdoctoral scholars: **11 completed, 4 current**

PhD's: **29 graduated** (including 2 IBM PhD Fellows, 1 Lam research PhD Fellow), **8 current**;

MS's: **9 graduated** (with thesis option)

Total number of issued United States patents: **186**

Graduated Students (29 PhDs, 11 Postdoctoral Researchers, 7 Masters)

1. Tanmoy Maiti (Postdoctoral Associate) (08/09-08/10) (Currently, Assistant Professor at Indian Institute of Technology, Kanpur, India)
2. Ramakrishnan Krishnan, PhD, 12/2009: Reliability Effects Of Soft Errors and NBTI in Current and Emerging Digital Circuits (Currently, Senior Staff Engineer, Advanced Technology Platforms Group, Taiwan Semiconductor and Manufacturing Corp (TSMC), Hsinchu, Taiwan)
3. Saurabh Mookerjee, PhD, 08/2010: Simulation, Design and Fabrication of Tunnel Transistors with steep sub-threshold slopes (Currently, Senior Device Engineer, Logic Technology Development, Intel Corporation, Hillsboro, Oregon)
4. Wei-Chieh Kao, MS (Thesis), 05/2010: Impact of Non-ideal Interfaces on Transistor Performance (Currently, PhD student at Arizona State University)
5. Vikram Sampat Kumar, MS (Thesis), 04/15/2010: An FPGA-based Real Time Tracking For Indoor Environment
6. Srijith Rajamohan, MS (Thesis), 04/2010: A Neural Network based Classifier on the Cell Broadband Engine
7. Ashkar Ali, MS (Thesis), 03/2009: Transport in Silicon Quantum Dots Embedded in a Rare Earth Oxide
8. Chad Ostrowski, BS (Honor's Thesis) 12/2009: Analytical Modeling of Tunnel Diodes
9. Vinay Saripalli, PhD, 10/31/11: Device Architecture Co-Design for Ultra Low Power Logic Using Emerging Tunneling Based Devices (Currently Senior CAD Engineer, Intel Corporation, Santa Clara)
10. Zhao Feng, PhD, 08/31/2011 : Ultra Sensitive Magnetic Sensors Integrating the Giant Magnetoelectric Effect with MEMS and Advanced CMOS (Currently Design Engineer, Texas Instruments, Dallas)
11. Salil Mujumdar, MS (Thesis), 05/2011: Strain Engineering in Nanoscale Transistors (Thesis option) (Current Device Engineer, Inter Molecular Foundry, San Jose)

12. Ashish Agrawal, MS (Thesis), 05/2011: Noise measurement and modeling of nanoscale devices (Thesis option) (Currently Ph.D. candidate at Penn State)
13. Ashkar Ali, PhD, 06/2012: Design and Fabrication of Ultra-low power and High Performance Quantum-well Transistors (IBM PhD Fellow 2010-2011, Currently Senior Device Engineer at Intel Corporation)
14. Feng Li, PhD, 08/2012: Ultra-sensitive Chip-Scale Magnetometers (Currently Design Engineer, Freescale Semiconductors)
15. Srinidhi Kestur, PhD, 01/2012: Accelerating computationally intensive applications using Reconfigurable systems (Currently Senior Design Engineer, Intel Corporation)
16. Euichul Hwang, PhD, 09/2012: Multi-gate III-V Metal Oxide Semiconductor FETs (Currently Device Engineer, Samsung Advanced Institute of Technology, SAIT)
17. Dheeraj Mohata, PhD, 01/2013: Arsenide-Antimonide Hetero-Junction Transistors for Low Power Logic Applications (Currently Integration Engineer, RF Micro Devices)
18. Ayan Kar, Post Doctoral Researcher, 02/13: (Currently Senior Device Engineer, Intel Corporation)
19. Eugene Freeman, MS (Thesis), 11/13: Correlated Electron Based Switches (Currently PhD student at Penn State)
20. Bijesh Rajamohanan: PhD, 05/2014: Fabrication, Characterization and Physics of III-V Tunneling Field Effect Transistors for Low Power Logic and RF Applications (Currently Senior Device Engineer, Sandisk Corporation)
21. Lu Liu: PhD, 05/2014: Classical and Coulomb Blockade III-V Multi-Gate Quantum Well Field Effect Transistors for Ultra Low Power Logic Applications (Currently Senior Device Engineer, Intel Corporation)
22. Ashish Agrawal: PhD, 12/2014: Physics and Technology of Strained Germanium Quantum Well FinFET for Low Power P-Channel Application (Currently Senior Device Engineer, Intel Corporation)
23. Huichu Liu: PhD, 5/2014: Circuit-Device Interaction for Steep Switching Slope Devices (Currently Senior Architecture Engineer, Intel Corporation)
24. Matt Hollander: PhD, 11/2015: Two-Dimensional Materials for Novel Electronic Applications: The Graphene Mixer and TaS₂ Hyper FET (Currently Senior Device Engineer, Micron Corporation)

25. Arun VT: PhD, 7/2015: Physics and technology of nanoscale III-V field effect transistors for low power electronics (Currently Senior Device Engineer, Intel Corporation)
26. Nidhi Agrawal: PhD, 7/2015: Numerical Simulation of Variation in 3D NonSilicon Transistors (Currently Senior Reliability Engineer, Micron Corporation)
27. Ali Razavieh (Post Doctoral Associate, PhD, Purdue University, West Lafayette, Indiana)(Currently, Senior Device Engineer, Global Foundries)
28. Bikas Das (Post Doctoral Associate, PhD, Indian Association of Cultivation of Science, Kolkata, India)(Currently Assistant Professor, IISER Calcutta)
29. Sandeepan Das Gupta (Post Doctoral Associate, PhD. Vanderbilt University)(Currently Senior Device Engineer, Micron Corporation)
30. Himanshu Madan: PhD, 12/2015: RF Electronics based on Emerging Devices (Currently Senior Device Engineer, Intel Corporation)
31. Mike Barth: Ph.D., 08/01/2016: Antimonide based Low Power Nanoelectronics (Currently Senior Process Engineer, Intel Corporation)
32. Rahul Pandey: Ph.D., 08/01/2016: Electrical Noise in Emerging Devices (Currently Senior Device Engineer, Intel Corporation)
33. Ramkrishna Ghosh (Post Doctoral Associate, PhD, Indian Institute of Science, IISc, Bangalore, India) (Assistant Professor, Jawaharlal Nehru University (JNU), Delhi, India)
34. Pankaj Sharma (Post Doctoral Associate, PhD, EPFL, Lausanne, Switzerland)(Currently Senior Process Integration Engineer, Micron Corporation)
35. Nikhil Shukla: PhD, 08/01/2017: Computational Device and Circuit Concepts using Electronic Phase Transition (Assistant Professor, ECE, University of Virginia, Charlottesville)
37. Xueqing Li (Post Doctoral Associate, PhD, Tsinghua University)(Assistant Professor, Tsinghua University, Beijing, China)
38. Matt Jerry: Ph.D.,10/01/2018: Collective Phenomena Based Solid State Devices and Their Natural Computing Applications (Senior Memory Array Engineer, Micron)

39. Kai Ni (Post Doctoral Associate, PhD, Vanderbilt University)(Assistant Professor, University of Notre Dame, Notre Dame, Indiana, USA)
40. Benjamin Grisafe: Ph.D., 07/23/2020: Threshold Switching Phenomena and Their Application to Electronic Devices (Process Integration Engineer, Northrop Grumman)
41. Jeff Smith: Ph.D., 11/20/2020: Novel Silicon and Non-Silicon Transistors For Low Power Logic Applications (Senior Device Engineer, Wolfspeed)
42. Abhishek Khanna: Ph.D., 4/6/2023: Phase Transition Oscillators Based Ising Machines for Combinatorial Optimization Problem Solvers (Senior Device Integration Engineer, Micron)
43. Wriddhi Chakrobarty: Ph.D. 4/4/2022: Cryogenic CMOS Technology and Monolithic 3D Integration for high Performance Computing (Senior Process Integration Engineer, Intel Corporation)
44. Jorge Gomez: Ph.D., 06/2021: Functional Oxide-Based Emerging Nanoelectronic Devices for Low Power Logi, Memory and High Frequency Applications (Research Scientist, Meta)
45. Huacheng Ye: Ph.D., 08/01/2022: Sequential Monolithic Three Dimensional Circuits (Senior Device Engineer, Western Digital)
46. Sourav Dutta (Post Doctoral Associate, PhD, Georgia Tech) (Currently, Assistant Professor, University of Texas, Dallas)
47. Navnidhi Upadhyay (Post Doctoral Associate, PhD, University of Massachusetts, Amherst) (Currently, R&D Engineer, EMD Electronics)
48. Khandker Akif Aabrar, PhD, 08/2024 Ferroelectric and Monolithic 3D (M3D) Memory Technologies for High-Performance, Energy-Efficient Computing First employment (Senior Process Integration Engineer, Micron Technology)

Current Research Interests

- Energy Efficient CMOS (e.g. next-gen gate stacks and source-drain contacts)
- Cryogenic FinFET CMOS (e.g. multi-state gain-cell memory)
- Monolithic 3D integration (e.g. low temperature processed oxide semiconductor transistors)
- In-memory computing (e.g. artificial neural networks using analog ferroelectric memory)
- Neuromorphic computing (e.g. spiking neural networks)
- Physics Inspired computing (e.g. Potts and Ising solvers using coupled oscillators)

- Biology Inspired computing (e.g. coupled oscillators using living heart muscle cells)

Awards and Honors

- SIA (Semiconductor Industry Association)-SRC (Semiconductor Research Corporation) University Researcher Award (2024)
- VLSI Technology and Circuits Test of Time Award (2023)
- Facebook Faculty Award (2020)
- Stinson Endowed Professorship (2019)
- Fellow of National Academy of Inventors (NAI) (2017) for “outstanding inventions that have made a tangible impact on quality of life, economic development and the welfare of society”
- Penn State Engineering Alumni Society (PSEAS) Premier Research Award (2015)
- IEEE Fellow for “contributions to for contributions to high-performance advanced silicon and compound semiconductor transistor technologies” (2013)
- SEMI Award for North America “in recognition of their pioneering work in the development, integration and introduction of a successful high-k dielectric and metal electrode gate stack for 45 nm CMOS IC production” (2012)
- IBM Faculty Award (2012)
- Penn State Engineering Alumni Society (PSEAS) Outstanding Research Award (2012)
- Distinguished Lecturer of IEEE Electron Devices Society (2011)
- Joseph Monkowsky Professorship for Faculty Early Career Development, The Pennsylvania State University (2007)
- Intel Achievement Award (the highest technical honor at Intel) for “developing the world’s first high-K/metal gate CMOS transistors with record-setting performance” (2003)
- Divisional Achievement Award from Intel Logic Technology Development Group for “invention and successful demonstration of high performance Tri-gate CMOS transistors” (2002)
- All India Rank of 124 among 300,000 students who appeared for Indian Institute of Technology Joint Entrance Examination (IIT-JEE) (1995)

Research Supervision (current)

Postdoctoral Researcher (5)

1. Sunbin Deng, PhD, Hong Kong University of Science & Technology (HKUST)
2. Hyeonwoo Park, PhD, Seoul National University (SNU)
3. Hyun Jae Lee, PhD, Seoul National University (SNU)
4. Dr. Songhyeon Kuk, PhD, Korea Advanced Institute of Technology (KAIST)
5. Dr. Junsu Yu, PhD, Seoul National University (SNU)

Doctoral Students (9)

1. Dyutimoy Chakraborty: Analog Compute in Memory (Start Date: 09/2023)
2. Rinku Dutta: Electro-thermal Transport in Extremely Scaled Transistors (Start Date: 09/2023)
3. Sharadindugopal Kirtania: Reliability Physics of Oxide Transistors (Start Date: 08/2022)
4. Eknath Sarkar: Gate-all-around Transistors (Start Date: 09/2023)
5. Jaewon Shin: Ferroelectric Non-volatile transistors (Start Date: 09/2023)
6. Chengyang Zhang: Zero Barrier Schottky Contact to Semiconductors (Start Date: 09/2022)
7. Emmanuel Quezada: BEOL Oxide Transistors (Start Date: 08/2024)
8. Ishaan Joshi: State Space Model Implementation on Ferroelectric Hardware (Start Date: 08/2025)
9. Wan-Hsuan Lo: DTCO for CFET (Start Date: 08/2025)

MS Students (0)

Publications

Book Chapters

- [1] V. Saripalli, V. Narayanan and S. Datta, “Ultra Low Energy Binary Diagram Circuits Using Few Electron Transistors”, *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, Springer Berlin Heidelberg, October 2009
- [2] V. Eachempati, R. Das, V. Narayanan, Y. Xie, S. Datta and C. Das, “HeTERO: Hybrid Topology Exploration for RF based On Chip Networks”, *Communication Architectures for System-on-Chip (SoC)*, CRC Press, September 2010
- [3] S. Datta, D. Schlom, “Gate Oxides beyond SiO₂”, *Multifunctional Oxide Heterostructures*, Oxford University Press, September 2010
- [4] S. Datta, “III-V MOSFETs”, *Future Intelligent Integrated Systems: New Paths to Augmented Silicon CMOS Technologies*, WSPC-Pan Stanford (Singapore), January 2013
- [5] Nikhil Shukla, S. Datta, A. Parihar, A. Raychowdhury, “Computing with Relaxation Oscillators”, *Future Trends in Microelectronics*, Wiley, March 2016
- [6] J. Robinson, M. Hollander, S. Datta, “Epitaxial Graphene: Progress on Synthesis and Device Integration”, *2D Materials for Nanoelectronics*, CRC Press, May 2016
- [7] Suman Datta, “Tri-Gate Transistors”, *Nanoscale Silicon Devices*, CRC Press, Jan 2016

- [8] Ahmedullah Aziz, Sumitha George, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, “Sensing in Ferroelectric Memories and Flip-Flops”, *Sensing of Non-Volatile Memory Demystified*, Springer, Aug 2018
- [9] Xueqing Li, Moon Seok Kim, Ahmedullah Aziz, Matthew Jerry, Nikhil Shukla, John Sampson, Sumeet Gupta, Suman Datta, Vijaykrishnan Narayanan, “Emerging Steep-Slope Devices and Circuits: Opportunities and Challenges”, *Beyond CMOS Technologies for Next Generation Computer Design*, Springer, Aug 2018
- [10] Ahmedullah Aziz, Sandeep Krishna Thirumala, Danni Wang, Sumitha George, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, “Sensing in Ferroelectric Memories and Flip-Flops, *Sensing of Non-Volatile Memory Demystified*, Springer, 2019
- [11] Robert Chau, Suman Datta, “Field-Effect Transistor R&D in the United States: Past, Present, and Future”, *The FET Centennial*, Wiley, 2026

Journal Articles

- [258] Madhav Vadlamani, Dyutimoy Chakraborty, Jianwei Jia, Halid Mulaosmanovic, S. Duenkel, S. Beyer, Suman Datta, S. Yu, “Cryogenic Characterization of Ferroelectric Nonvolatile Capacitors” in IEEE Transactions on Electron Devices, Mar 2026
- [257] J. Sonawane, S. Deng, C. Zhang; Omkar Phadke; Faaiq Waqar; Suman Datta "Asymmetric Underlap–Overlap 2T Gain Cell Enabling Voltage-Sensing Readout With Minimized Capacitive Coupling," in IEEE Electron Device Letters, vol. 47, no. 3, pp. 522-525, March 2026
- [256] X. Niu, S. Deng, F. X. Liang, K. A. Aabrar, Y. Qin, J. Duan, Suman Datta, V. Narayanan, X. Hu, K. Ni, "Forming Process in BEOL Oxide Channel Ferroelectric FET," in IEEE Transactions on Electron Devices, vol. 72, no. 12, pp. 6963-6969, Dec. 2025.
- [255] K. Toprasertpong, T. Mikolajick, Suman Datta, Q. Huang, Shinichi Takagi, “Emerging applications: Neuromorphic computing and reservoir computing,” MRS Bulletin, vol. 50, no. 9, pp. 1032-1042, 2025.
- [254] J. Kwak, S. Deng, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, J. Lee, Suman Datta, S. Yu, “Machine learning-assisted compact modeling of AC stress-induced bias temperature instability in top- and bottom-gate W-doped In₂O₃ channel semiconductor transistors,” IEEE Transactions on Electron Devices, vol. 72, no. 8, pp. 4163-4169, 2025.

- [253] J. Kwak, S. Deng, J. Lee, Suman Datta, S. Yu, "Backside active power delivery with hybrid DC-DC converter enabled by amorphous oxide semiconductor transistors," *IEEE Transactions on VLSI Systems*, vol. 33, no. 8, pp. 2153-2162, 2025.
- [252] O. Phadke, P. V. Ravindran, H. Mulaosmanovic, S. Dunkel, S. Beyer, A. I. Khan, Suman Datta, S. Yu, "Impact of the endurance measurement methodology on ferroelectric field effect transistor under the capacitive non-destructive read," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2312-2318, 2025.
- [251] O. Phadke, K. A. Aabrar, G. Choe, Y.-C. Luo, S. G. Kirtania, A. I. Khan, Suman Datta, S. Yu, "Effect of DC stress on low-frequency noise characteristics of W-doped In₂O₃ BEOL transistors," *IEEE Transactions on Electron Devices*, vol. 72, no. 3, pp. 1489-1493, 2025.
- [250] H. Park, S. G. Kirtania, E. Sarkar, D. Chakraborty, C. Zhang, H. J. Lee, J. Shin, S. Yu, A. I. Khan, H. Kim, C. Im, M. J. Hong, D. Ha, Suman Datta, "Extreme EOT scaling in tungsten-doped In₂O₃ MOSFETs for enhanced stability and drive current," *IEEE Transactions on Electron Devices*, vol. 72, no. 12, pp. 7136-7144, 2026.
- [249] S. G. Kirtania, H. Park, O. Phadke, E. Sarkar, D. Chakraborty, F. G. Waqar, J. Shin, A. I. Khan, S. Yu, Suman Datta, "Amorphous indium oxide channel FeFETs with write voltage of 0.9 V and endurance > 10¹² for refresh-free embedded memory," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2691-2699, 2025.
- [248] E. Sarkar, C. Zhang, D. Chakraborty, S. G. Kirtania, K. A. Aabrar, H. Park, J. Shin, H. J. Lee, M. Tian, A. I. Khan, S. Yu, Suman Datta, "First demonstration of high-performance and extremely stable W-doped In₂O₃ gate-all-around (GAA) nanosheet FET," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2662-2669, 2025.
- [247] S. Woo, G. Choe, A. I. Khan, Suman Datta, S. Yu, "Design of superlattice ferroelectric-metal field-effect transistor for triple-level cell 3D NAND Flash," *Microelectronic Engineering*, vol. 295, pp. 112276, 2025.
- [246] L. Fernandes, P. V. Ravindran, J. Chen, M. Tian, D. Das, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, Suman Datta, A. I. Khan, "Optimizing memory window for ferroelectric NAND applications: An experimental study on dielectric material selection and layer positioning," *IEEE Transactions on Electron Devices*, vol. 72, no. 1, pp. 234-239, 2025.

- [245] P. Venkatesan, L. Fernandes, P. Ravikumar, C. Park, H. Tran, Z. Wang, H. Jayasankar, A. Garlapati, T. Song, M. Tian, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, S. Yu, Suman Datta, W. Kim, D. Ha, A. I. Khan, "Demonstration of robust retention in band engineered FeFETs for NAND storage applications using tunnel dielectric layer," IEEE Electron Device Letters, vol. 46, no. 3, pp. 397-400, 2025, editor's pick.
- [244] P. Ravikumar, P. Venkatesan, C. Park, N. Afroze, M. Tian, W. Chern, Suman Datta, S. Yu, S. Mahapatra, A. I. Khan, "The first switch effect in ferroelectric field-effect transistors," IEEE Transactions on Device and Materials Reliability, vol. 25, no. 3, pp. 365-370, 2025.
- [243] Tanvir H Pantha, A. Khanna, H. Ye, S. Niazi, M. P Kamal, B. Chakraborty, S. Alam, E. Weinstock, N. Babu, S. Mukhopadhyay, Y. Chiu, Suman Datta, K. Y Camsari, S. Dutta, "Addressing the Connectivity Bottleneck With BEOL FeFETs for 3-D CMOS + X Ising Machines," in IEEE Transactions on Electron Devices, vol. 72, no. 6, pp. 3335-3342, June 2025
- [242] S. G. Kirtania, H. Park, O. Phadke, E. Sarkar, D. Chakraborty, F. G Waqar, J. Shin, A. Khan, S. Yu, Suman Datta, "Amorphous Indium Oxide Channel FEFETs With Write Voltage of 0.9 V and Endurance >10¹² for Refresh-Free Embedded Memory," in IEEE Transactions on Electron Devices, vol. 72, no. 5, pp. 2691-2699, May 2025
- [241] FVE Hensling, P Vogt, J Park, SL Shang, H Ye, YM Wu, Suman Datta, D. Schlom, "Fully Transparent Epitaxial Oxide Thin-Film Transistor Fabricated at Back-End-of-Line Temperature by Suboxide Molecular-Beam Epitaxy", Advanced Electronic Materials, vol. 11, no. 3, pp. 24004992025, 2025
- [240] P Venkatesan, L Fernandes, P Ravikumar, C Park, H Tran, Z Wang, S Lim, K. Kim, S. Yu, S. Datta, W. Kim, D. Ha, A. Khan, Demonstration of Robust Retention in Band engineered FEFETs for NAND Storage Applications using Tunnel Dielectric Layer, IEEE Electron Device Letters, 2024.
- [239] L Fernandes, PV Ravindran, J Chen, M Tian, D Das, H Chen, W Chern, ... S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, S. Datta, A. Khan, Optimizing Memory Window for Ferroelectric Nand Applications: An Experimental Study on Dielectric Material Selection and Layer Positioning, IEEE Transactions on Electron Devices, vol. 72, no. 1, pp. 234-239, Jan. 2025
- [238] DB Durham, M Noor, KA Aabrar, Y Liu, S. Datta, K Cho, S Guha, C Phatak Direct Imaging of Asymmetric Interfaces and Electrostatic Potentials inside a Hafnia–Zirconia Ferroelectric Nanocapacitor, ACS Applied Materials & Interfaces 16 (49), 68562-68568, 2024

- [237] S. Deng, J. Kwak, J. Lee, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K.A. Aabrar, S. Yu, S. Datta, "Monolithic Switched-Capacitor DC-DC Converters Using BEOL-Compatible Oxide Power Transistors and Superlattice MIM Capacitors," in IEEE Transactions on Electron Devices, vol. 71, no. 12, pp. 7999-8006, Dec. 2024
- [236] Z. Zhao, S. Woo, K. A. Aabrar, S. G. Kirtania, Z. Jiang, S. Deng, Y. Xiao, H. Mulaosmanovic, S. Duenkel, D. Kleimaier, S. Soss, S. Beyer, R. Joshi, S. Meninger, M. Mohamed, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, V. Narayanan, S. Datta, S. Yu, K. Ni, Paving the Way for Pass Disturb-Free Vertical NAND Storage via a Dedicated and String-Compatible Pass Gate, ACS Applied Materials & Interfaces, vol 16, No 41, pp 55619-55626, 2024
- [234] O. Phadke, S. Gopal Kirtania, D. Chakraborty, S. Datta and S. Yu, "Suppressed Capacitive Coupling in 2 Transistor Gain Cell with Oxide Channel and Split Gate," in IEEE Transactions on Electron Devices, vol. 71, no. 11, pp. 6749-6755, Nov. 2024
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- [25] S. Mookerjee* and S. Datta, "Comparative Study of Si, Ge and InAs Based Steep Subthreshold Slope Tunnel Transistors for 0.25V Supply Voltage Logic Applications," *IEEE Device Research Conference (DRC)*, pp. 47-48, Santa Barbara, California, June 2008. (First author supervised by the candidate)
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Recent Invited Talks (selected)

Universities

“Steep Slope Phase Transition FETs and their applications” École Polytechnique Fédérale de Lausanne EPFL, Lausanne, Switzerland, 1/2016

“Inter-band Tunnel Transistors: Opportunities and Challenges”, *NEEDS Seminar*, Purdue University, West Lafayette, IN, 10/2015

“Negative Capacitance Ferroelectric Transistors: A Promising Steep Slope Device Candidate?”, *NEEDS Seminar*, Purdue University, West Lafayette, IN, 10/2015

“Ultra Low Power Devices”, *IEEE EDS Distinguished lecture series in Workshop on Connected, Autonomously Powered Systems*, Columbia University, New York, NY, 04/2014

“Cool Device Strategies for Beyond CMOS Nanoelectronics”, *ECE Colloquium*, University of Texas, Dallas, TX, 02/2014

“Cool Device Strategies for Beyond CMOS Nanoelectronics”, *ECE Colloquium*, Cornell University, Ithaca, NY, 02/2012

“Nanoelectronics for Future Energy Efficient Information Processing”, *ECE Colloquium*, University of Illinois, Urbana Champagne, IL, 03/2012

“Tunnel Transistor Based Energy Efficient Logic”, *IEEE EDS Distinguished lecture series in Electronics/Photonics*, Ohio State University, Columbus, OH, 04/2010

“Energy Efficient Logic Transistors using Compound Semiconductors” Cornell University Electron Devices Society Lecture Series, Cornell University, Ithaca, NY, 04/2010

“Logic and Memory Design using Inter-band Tunnel Transistor” Nanoseminar Seminar Series, Arizona State University, 03/2010

“Compound Semiconductor based Logic Elements” IEEE Electron Devices Mini Colloquium, Indian Institute of Technology, Mumbai, 01/2010

“Ushering in the Green Transistor Era”, Rochester Institute of Technology, Rochester, New York, 5/2009

“Green Transistors to Green Architectures”, Institut für Materialien und Bauelemente der Elektronik, Leibniz Universität Hannover (University of Hannover), Hannover, Germany, 10/2009

“Green Nanoelectronic Computing Devices”, as part of the annual workshop on Emerging Trends in Photonic and Electronic Device Research held sponsored by The University of Illinois chapters of the Optical Society of America (OSA) and the IEEE Electron Devices Society (EDS) in conjunction with the Micro and Nanotechnology Laboratory (MNTL), University of Illinois, Urbana Champagne, Illinois, 09/2008

“Recent Advances in Silicon and Non-Silicon Nanoelectronic Devices for High-Performance, Energy Efficient Logic Applications”, Penn State University Computer Science and Engineering Colloquia Series, 11/2007

“Emerging Nanoelectronic Devices for High-Speed and Ultra-Low Power Applications”, sponsored by the University of Wisconsin, Madison, Materials Research Science and Engineering Center (MRSEC) in association with Electrical Engineering Department University of Wisconsin, Madison, Wisconsin, 12/2006

“Ultra Low Power Nanoelectronics for the Logic technology”, Taipei Local Chapter of IEEE Electron Devices Society (EDS), National Tshao-tung University (NCTU), Hsinshu, Taiwan:, 12/2006

“Emerging Nanoelectronic Devices for High-Speed and Ultra-Low Power Application”, Electrical Engineering Colloquium, University of Texas, Austin, 3/2006

“Silicon Nano-Transistors and Nanotechnology for High-Performance Logic Applications”, sponsored by the IEEE Phoenix Section Components, Packaging, and Manufacturing Technology Society Chapter, & Waves and Devices Chapter, Arizona State University, Tempe, Arizona, 11/2003

Government/Industry

“Steep Slope Transistors”, Speaker at the IEEE Rebooting Computing Workshop, Washington DC, 10/12/2015

“Steep Slope Transduction FETs”, Invited Speaker at Global Foundries, Malta, New York, 10/21/2015

“Strained Germanium Quantum Well FinFETs”, Invited Speaker at Global Foundries, Malta, New York, 09/01/2015

“Steep Slope Transistors”, Keynote Speaker at the NSF sponsored “The Workshop for Energy Efficient Computing” Arlington, VA, 04/14/2015

“Function Stacks for Logic and Memory Devices”, Invited Speaker at the SEMATECH's 7th International Symposium on Advanced Gate Stack Technology, in Albany, New York, 09/29/2010-10/01/ 2010

“Non silicon logic elements for extreme voltage scaling “, Invited Speaker at the IBM MRC Workshop on III/V Devices IBM Research, Zurich, Switzerland, 09/2010

“Binary Decision Diagram Logic for Single Electron Devices and Tunnel FETs”, Invited Speaker at the Nanoelectronics Research Initiative (NRI) sponsored Architecture & Device Benchmarking Workshop, University of Notre Dame, 08/09/2010

“Tunnel Transistors: From Circuits to Architecture”, Invited Speaker at the Nanoelectronics Research Initiative (NRI) sponsored Architecture & Device Benchmarking Workshop, University of Notre Dame, 08/09/2010

“High mobility channel MOSFETs: to include or not to include in the ITRS?”, Panelist at the Sematech/IMEC III-V Workshop for discussion on inclusion high mobility channel MOSFETs in the ITRS, Hilton Hawaiian Village, Honolulu, 06/2010

“Green Transistors to Green Architectures”, Tutorial at the 16th IEEE International Symposium on High-Performance Computer Architecture (HPCA), Bangalore, India, 01/2010

“High Mobility Channel MOSFETs”, Panelist at the Workshop for Future III-V Complementary Metal–Oxide–Semiconductor (CMOS) Technology, Washington DC , December 2009

“Heterojunction Tunnel Transistor Logic,” Intel on-site NRI sponsored PI’s Workshop, Intel Corporation, Portland, Oregon, 8/2009

“Tunnel Transistor Logic”, Intel Microprocessor Research Lab Seminar, Portland, Oregon, 10/09

"Looking Beyond Silicon - A Pipe Dream or the Inevitable Next Step?" Panelist on the IEDM sponsored evening panel called (This panel assembled internationally recognized panelists to discuss the future of Complementary Metal-Oxide Semiconductor (CMOS) and beyond CMOS for leading-edge advanced integrated circuit applications", 12/2007

"III-V Complementary Metal-Oxide Semiconductor (CMOS) on Si: Technical and Manufacturing Needs" Panelist on the Sematech and Aixtron sponsored workshop on readiness of III-V MOSFET Technology. This workshop received world-wide press coverage under the heading “III-V Compounds Emerging as Prime Materials for Future NMOS Channels, Technologists Indicate at SEMATECH & AIXTRON Workshop,” Washington, D.C., 12/2007

Inventions

[186] US Patent # 11532355 “Non-volatile multi-level cell memory using a ferroelectric superlattice and related systems”

[185] US Patent # 11031482 “Gate electrode having a capping layer”

[184] US Patent # 10937907 “Method for fabricating transistor with thinned channel ”

[183] US Patent # 10839880 “Low power sense amplifier based on phase transition material”

[182] US Patent # 10707319 “Gate electrode having a capping layer”

[181] US Patent # 10672475 “Nonvolatile digital computing with ferroelectric FET”

[184] US Patent # 10475514 “Nonvolatile digital computing with ferroelectric FET”

[183] US Patent # 10367093 “Method for fabricating transistor with thinned channel”

[182] US Patent # 10262714 “Low power sense amplifier based on phase transition material”

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Professional Activities

Associate Editor, IEEE Journal on Exploratory Solid-State Computational Devices and Circuits

Distinguished Lecturer, Electron Devices Society, IEEE

Fellow of IEEE; **Fellow** of National Academy of Inventors

General Chair, *IEEE International Electron Devices Meeting (IEDM '20)*, San Francisco, December 2020

Chair, *Technical Program, IEEE International Electron Devices Meeting (IEDM '19)*, San Francisco, December 2019

Vice Chair, *Technical Program, IEEE International Electron Devices Meeting (IEDM '18)*, San Francisco, December 2018

Chair, *Short Course, IEEE International Electron Devices Meeting (IEDM '17)*, San Francisco, December 2017

Vice Chair, *Short Course, IEEE International Electron Devices Meeting (IEDM '16)*, San Francisco, December 2016

Chair, *Publications, IEEE International Electron Devices Meeting (IEDM '15)*, Washington DC, December 2015

TPC Member, Focus Sessions Chair, VLSI Technology Symposium (*VLSI Symposium '23*), Honolulu, Hawaii

TPC Member, Focus Sessions Chair, VLSI Technology Symposium (*VLSI Symposium '22*), Kyoto, Japan

TPC Member, Forum Chair, VLSI Technology Symposium (*VLSI Symposium '21*), Honolulu, Hawaii

TPC Member, Arrangements Chair, VLSI Technology Symposium (*VLSI Symposium '20*), Kyoto, Japan

TPC Member, Arrangements Chair, VLSI Technology Symposium (*VLSI Symposium '19*), Honolulu, Hawaii

TPC Member, VLSI Technology Symposium (*VLSI Symposium '18*), Kyoto, Japan

TPC Member, Design Automation Conference (DAC '16, DAC '17)

TPC Member, International Conference on Simulation of Semiconductor Processes & Devices (SISPAD '16), Dresden, Germany

TPC Member, International Reliability Physics Symposium (IRPS '16, IRPS '17)

Chair, *Publicity, IEEE International Electron Devices Meeting (IEDM '14)*, San Francisco, California, December 2014

Chair, *Tutorials, IEEE International Electron Devices Meeting (IEDM '12)*, San Francisco, California, December 2012

General Chair, *IEEE Device Research Conference (DRC)*, Notre Dame University, University Park, PA, June 2013

Program Chair, *IEEE Device Research Conference (DRC)*, Pennsylvania State University, University Park, PA, June 2012

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Chair, *Emerging Technologies, IEEE International Electron Devices Meeting (IEDM '11)*, Washington DC, December 2011

TPC Vice-Chair, *IEEE Device Research Conference (DRC)*, University of California, Santa Barbara, CA, June 2011

Chair, *Quantum, Power, and Compound Semiconductor Devices Sub-Committee, IEEE International Electron Devices Meeting (IEDM '10)*, San Francisco, California, December 2010

Organizer and Chair, *Rump Session on Embedded Memory called "Looking for Extra Cache", IEEE Device Research Conference (DRC '10)*, Univ. of Notre Dame, June 2010

Organizer and Chair, *Rump Session on Steep Slope Transistors called "Steep Slope or Slippery Slope", IEEE Device Research Conference (DRC '10)*, Penn State University, June 2009

TPC Member, *IEEE Device Research Conference (DRC '10)*, Univ. of Notre Dame, June 2010

TPC Member, *Electronic Materials Conference (EMC '10)*, Univ. of Notre Dame, June 2010

TPC Member, *Semiconductor Interface Specialist Conference (SISC '10)*, San Diego, California, December 2010

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Seville, Spain, September, 2010

TPC Member, *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED '10)*, Austin, Texas, August 2009

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Athens, Greece, September, 2009

TPC Member, *Quantum, Power, and Compound Semiconductor Devices, IEEE International Electron Devices Meeting (IEDM '09)*, Baltimore, Maryland, December 7–9, 2009

TPC Member, *IEEE Device Research Conference (DRC '09)*, Penn State Univ., June 2009

TPC Member, *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED '09)*, San Francisco, California, August 2009

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Edinburgh, Scotland, September 2008

TPC Member, *Silicon Nanoelectronics Workshop (SNW '08)*, Honolulu, Hawaii, June 2008

TPC Member, *International Symposium on VLSI Technology, Systems, and Applications (2005 IEEE VLSI-TSA)*, Hinshu, Taiwan, April 25-27, 2005

Arrangements Chair, *IEEE Device Research Conference (DRC)*, Penn State University, June 2009

Panel Chair, *ACM/IEEE International Symposium Low Power Electronics and Design (ISLPED '09)*, San Francisco, California, August 2009

Session Chair, “*Non Volatile Memory*” at the *High-K Dielectric Materials and Gate Stack Symposium, Electrochemical Society Meeting (ECS '09)*, Vienna, Austria, October 2009

Session Chair, “*MOS Devices*” at the *IEEE Device Research Conference (DRC '09)*, June 2009

Session Chair, “*MOS Devices*” at the *IEEE Device Research Conference (DRC '09)*, June 2009

Session Co-Chair, “*III-V Logic Transistors with Advanced Gate Stack*” at the *IEEE International Electron Devices Meeting (IEDM '09)*, Baltimore, Maryland, December 8, 2009

Session Co-Chair, “*Heterostructure High-Speed Devices*” at the *IEEE International Electron Devices Meeting (IEDM '08)*, San Francisco, California, December 2008

Session Chair, “*Materials for Ge and Si devices*” at the *International Conference on Insulating Films on Semiconductors (INFOS '07)*, Athens, Greece, June 2007

Member, “*Front End Processes*” Tab and contributor to the 2003 Version of the *International Technology Roadmap for Semiconductors (ITRS)*

Member, *American Society for Engineering Education*, 2007 – 2015

NSF Review Panelist- In EPDT, EMT programs in years 2008-2017

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ACM Journal on Emerging Technologies in Computing Systems

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Applied Physics Letters

Journal of Applied Physics

Journal of Vacuum Science and Technology B

Journal of Nanotechnology

Nanotechnology (Journal) from Institute of Physics (IOP)

Journal of Electronic Materials

ACS Nano

IEDM, DRC, ESSDERC, VLSI-TSA, IEEE ISLPED, SISC, EMC, IEEE SNW, VLSI Symposia
on Technology and Circuits